

Replacing Gaussian observations in memory-constrained inference

OpenAI

September 27, 2026

Abstract

Replacing the Gaussian rows used to select a finite message by independent rows increases the remaining conditional information by at most Cd , for a uniform spherical signal, message entropy at most d^2 , and the specified row dimensions proportional to d . As an application, we prove that learners with $M = o(d^2)$ persistent bits need $T = \Omega(d \log(1/\epsilon))$ exact observations to attain uniform-sphere angular success at least $3/5$, for $0 < \epsilon \leq 1/10$ and a deterministic finite horizon.

Contents

1	Introduction	2
2	The two experiments and their information difference	5
3	A projection moment with shared and fresh rows	10
4	A critical radius makes the density terms cancel	13
5	Posterior density levels and auxiliary projections	16
6	Actual and fresh projections through independent kernels	25
7	A finite Gaussian-fiber comparison	34
8	From row comparisons to streaming lower bounds	42

1 Introduction

A finite message chosen from data changes the conditional distribution of those data. This elementary fact becomes a central issue in memory-constrained inference. Suppose a signal S is observed through exact Gaussian equations and a learner retains a finite state W . After conditioning on W , the rows used to select that state are generally biased. An independent Gaussian matrix has the same unconditional row law, but it need not leave the same information about S once its labels are revealed.

This paper quantifies that change. We compare information beyond a wholly independent auxiliary projection with information beyond a projection containing the actual rows used by the learner. The two experiments keep the signal–message marginal and the signal–matrix marginal fixed. Their full joint laws differ. At suitable row dimensions proportional to d , the comparison cost is $O(d)$ when the message entropy is at most d^2 . We also prove the projection moment that makes this possible and develop two further comparisons based on replacing one row at a time and on a two-signal fiber measure.

Here is one precise comparison. Write H for discrete entropy in nats, h for differential entropy, and $I(\cdot; \cdot | \cdot)$ for conditional mutual information. Let S be uniform on S^{d-1} , put $m = \lfloor d/10 \rfloor$ and $\ell = \lfloor d/2 \rfloor$, and let A be an m -row standard Gaussian matrix independent of S . A countable message W may have arbitrary joint dependence on (S, A) , subject to $H(W) \leq d^2$. Draw an $(\ell - m)$ -row Gaussian matrix C independently of (S, A, W) , and an ℓ -row Gaussian matrix G independently of (S, W) . Theorem 4.1 proves

$$I(S; W | G, GS) \leq I(S; W | A, C, AS, CS) + Kd$$

for all sufficiently large d , with an absolute threshold and an absolute constant K . Thus replacing rows that helped select the message by fresh rows costs only $O(d)$ additional conditional information, despite the possible bias of A given W .

Memory, exact equations, and prior methods

Memory restrictions affect statistical estimation even when computation is unrestricted. Steinhardt and Duchi established memory-dependent minimax bounds for sparse noisy regression [10]. Raz’s branching-program lower bounds for parity learning and broader discrete learning tasks showed how finite width can force a large sample cost [6, 7]. Sharan, Sidford, and Valiant developed a closely related finite-state framework for continuous regression [9].

Their precision theorem uses isotropic Gaussian covariates with independent uniform additive noise of half-width $2^{-d/5}$. For Euclidean accuracy d^{-r} , with $r \leq O(d/\log d)$, at most $d^2/4$ bits of memory, and success probability at least $2/3$, it gives an $\Omega(d \log r)$ sample lower bound [9, Theorem 1]. The present observation law has exact labels, which contain more information than noisy labels. Accordingly, our proof needs an estimate for exact projection densities. Sharan, Sidford, and Valiant’s expansion of a projection moment into independent signal copies is a direct method precedent [9, Section 7, Lemma 12]; their one-row interval estimate uses a positive noise scale and global L^2 control, while the mixed multirow estimate below is proved from local mass bounds. Dagan, Kur, and Shamir give related space lower bounds for an approximate-nullvector task with independent Gaussian rows and for empirical regression on a randomly ordered finite data set with exactly consistent labels [2, Theorems 3 and 10]. The latter result bounds the empirical residual under a hard joint data distribution; our recovery problem uses independent Gaussian rows and a fixed signal.

Information-theoretic analyses of adaptive selection also compare a selected statistic with an independent reference. Russo and Zou bound the divergence between a selected statistic on the observed data and its evaluation on an independent copy by the information used for selection [8, Section 2.4, Proposition 4]. Xu and Raginsky bound an expectation difference between a joint law and the product of its marginals for a test that is sub-Gaussian under the product law [12, Section 3.1, Lemma 1]. These comparisons explain the role of an information charge for selection. Here the test is an exact projection density, and a locally proved q th moment makes the row-information charge appear with coefficient $1/q$.

The geometric calculation also belongs to the tradition of controlling projection energies by inverse distances. Mattila’s two-point averaging estimate gives a classical instance of that principle [4, proof of Lemma 5.1, p. 236]. Our mixed moment contains both inverse lengths and inverse distances to the span of earlier signal differences. The specific Gaussian factors, dimension conditions, and finite fiber normalization are proved in this paper.

The streaming model and the application

The comparison theorems concern finite messages with arbitrary dependence on a signal and some Gaussian rows. Their streaming application uses the following model.

Definition 1.1 (Finite persistent-state learner). Fix $d \geq 2$, an integer memory bound $M \geq 0$, an angular accuracy $0 < \epsilon \leq 1/10$, and a deterministic finite integer horizon $T \geq 0$. The signal is a unit vector. It may be fixed before sampling, or sampled from a specified prior; the average-case statements in this paper use uniform probability on S^{d-1} . Pre-generate independent

$$X_t \sim N(0, I_d), \quad Y_t = \langle X_t, S \rangle, \quad t \geq 1.$$

The row sequence is independent of S .

Let Ξ be a shared rule seed, constant if no shared seed is used, and let U_0 be the initial finite state. The pair (Ξ, U_0) is jointly independent of $(S, (X_t)_{t \geq 1})$; Ξ and U_0 may depend on one another. The original randomized experiment is jointly measurable in the signal, the complete row sequence, the shared seed, initialization, and all transition and output randomness. All variables and kernels are taken on standard Borel spaces, with the completed-measurability convention below for sample arguments.

At every index, including index zero, the persistent state has at most 2^M possible values. Conditional on Ξ , the rule at time t uses only the current state, the index t , and the entire current pair (X_t, Y_t) , together with fresh randomness that has no additional signal or sample information. The rule may perform unrestricted computation on that pair, but everything retained for later samples must be encoded in the new finite state. The learner receives the rows once in their prescribed order and cannot choose a row or revisit an earlier one.

The learner stops at an index $\tau \in \{0, \dots, T\}$. Conditional on the fixed seed, its unit-vector output $\hat{S}$ is drawn from a kernel depending only on the terminal state and τ , using fresh output randomness. In particular the last observed pair can affect the output only through that terminal state. The seed may select nonuniform rules depending on d, ϵ, t , but it has no data-dependent content. Before the first sample, a stopping choice may use only the seed, the initial state, and fresh randomness.

For each fixed seed, state, and index, the transition and stopping probabilities may be Borel functions of the sample pair, or measurable in the completion of Gaussian row measure times Lebesgue label measure. This per-seed convention supplements the required joint measurability of the original mixed experiment; it does not replace it. Success means $\arccos \langle \hat{S}, S \rangle \leq \epsilon$.

The joint condition on (Ξ, U_0) ensures that fixing the rule seed leaves initialization data-independent. We record this reduction because the information potentials start from the conditional initial-state law.

Lemma 1.2 (Fixing the shared rule seed). *In the model above, for almost every seed value ξ ,*

$$\mathcal{L}(S, (X_t)_{t \geq 1}, U_0 \mid \Xi = \xi) = \mathcal{L}(S, (X_t)_{t \geq 1}) \otimes \mathcal{L}(U_0 \mid \Xi = \xi). \quad (1)$$

If the original uniform-prior success probability is at least p_0 , one may choose such a seed value whose conditional uniform-prior success is at least p_0 , with the same state bound and horizon. This assertion is about the uniform-prior experiment.

Proof. Disintegrate the joint independence of (Ξ, U_0) and $(S, (X_t)_{t \geq 1})$ over Ξ . It gives (1) for almost every seed. Joint measurability of the original experiment makes its conditional success probability a measurable function of ξ , whose mean is the original success probability. Averaging therefore supplies a seed with the stated success and factorization. The conditional initial-state law is retained. A later Borel replacement is performed separately in this fixed-seed experiment; no jointly measurable selection of replacement versions is required. $\square$

The critical-radius comparison gives the following streaming consequence: for $M(d) = o(d^2)$, every accuracy sequence $0 < \epsilon(d) \leq 1/10$, and uniform-sphere average success at least $3/5$, the deterministic horizon must satisfy

$$T(d) = \Omega\left(d \log \frac{1}{\epsilon(d)}\right).$$

The lower-bound constant is absolute; the eventual dimension threshold may depend on the memory sequence. Section 8 proves this statement and records the distinct constants and thresholds arising from the other comparisons. A learner that succeeds with probability at least $2/3$ for every fixed signal also has that success under the uniform prior, so the worst-signal consequence follows separately. The auxiliary projections used in the proof are analysis variables; the learner receives only the exact stream in Definition 1.1.

How the comparisons work

Section 2 defines the aligned and independent experiments explicitly. Their information difference contains a negative conditional row-information term. Data processing compares that term with the divergence between the actual conditional label law and the projection of the posterior $P_{S|W}$. What remains is the entropy of the independent projection plus an aligned log density.

Section 3 proves the analytic estimate for that density. In a q th moment, the actual rows are shared by all q factors while independent auxiliary rows are averaged separately in each factor. Gaussian integration produces inverse lengths and the inverse Gram determinant of the signal differences. A local mass bound controls thin tubes around their successive spans, and a second bound controls the sum over small radial scales. The section treats exact density versions and the order of the change of row law and limiting operation.

Sections 4 and 5 use this estimate after refining a posterior by its local mass. The critical-radius refinement records both a radius and its logarithmic mass level. Two further comparisons use a single maximal level or a pair of density labels. Their entropy costs are explicit, and the density-level terms cancel with opposite signs. The two-label argument also retains its own Gaussian-mollifier and first-density passage.

These comparisons have different scopes. The critical-radius result allows countable messages and $\lfloor d/10 \rfloor$ actual rows. The two-label result treats finite messages with the larger actual block $\lfloor d/8 \rfloor$; its Gaussian-mollifier density version supports an estimate at every fixed target. The one-level result uses only one refinement label, a net-weight entropy estimate, and $\lfloor d/32 \rfloor$ actual rows. It keeps the entropy-dependent comparison cost explicit rather than imposing $H(W) \leq d^2$ in its statement.

The last two comparisons use different geometric objects. Section 6 keeps the actual ending state fixed while replacing an actual prefix of rows by an independent suffix, one row at a time. Independent random kernels and a normalized reciprocal-distance selection control the successive changes. Section 7 constructs a finite measure on pairs having the same Gaussian label. It proves that measure's exact kernel and normalizes the reference before applying probability data processing. Local-mass ratios then bound the resulting information difference. The hybrid argument averages over the number of auxiliary observations, whereas the fiber comparison concerns a state transition through an actual block, rather than an arbitrary message jointly distributed with its rows. These scopes explain the separate applications below.

Finally, Section 8 applies these comparisons to successive fresh blocks. It includes a local Borel-version reduction for the fixed-seed uniform-prior experiment, the state and stopping-index costs, and the residual-sphere information required by an accurate output. A separate memory-unrestricted sample bound absorbs the final-block rounding cost. This separates the analytic comparisons from the bookkeeping needed to apply them to a learner.

2 The two experiments and their information difference

The comparison begins with two experiments that give the signal and the finite message the same joint distribution. They differ in the rows revealed alongside that message. Before defining them, we fix the measure and entropy conventions needed to interpret their exact labels.

Write σ for uniform probability on S^{d-1} , put $n = d - 1$, and let $B(z, t)$ be the closed Euclidean ball with center z and radius t . The notation γ_a denotes the law of an $a \times d$ matrix with independent $N(0, 1)$ entries. All logarithms are natural. Constants denoted by C are positive and absolute and may increase between occurrences.

Lemma 2.1 (Spherical caps and ambient balls). *On the unit sphere in $\mathbb{R}^a$, $a \geq 2$, a cap of angular radius $0 \leq \theta \leq \pi$ has uniform probability at most θ^{a-1} . On S^{d-1} ,*

$$\sigma(B(s, t)) \leq t^n \quad (s \in S^{d-1}), \quad \sigma(B(z, t)) \leq (2t)^n \quad (z \in \mathbb{R}^d), \quad t > 0.$$

Consequently a fixed unit-vector output, or a random unit-vector output independent of the signal, succeeds to angular error ϵ on a set of σ -mass at most ϵ^n .

Proof. The cap probability is the ratio of the integrals of $\sin^{a-2} u$ over $[0, \theta]$ and $[0, \pi]$. The numerator is at most $\theta^{a-1}/(a-1)$. Symmetry and $\sin u \geq 2u/\pi$ on $[0, \pi/2]$ bound the denominator below by $\pi/(a-1)$, proving the angular estimate.

For the centered Euclidean estimate it suffices to consider $t \leq 1$. Rotate s to the north pole. Points in $B(s, t)$ have last coordinate at least $1 - t^2/2 \geq 1/2$. As a graph above the first n coordinates, this part of the sphere has area element at most twice Lebesgue measure and projects into the n -ball of radius t . The entire sphere has area at least twice the volume of the unit n -ball, by projecting the two hemispheres. Their ratio is at most t^n . For $t > 1$, total mass one suffices. An ambient ball meeting the sphere is contained, on the sphere, in the ball of radius $2t$ about any one point of that intersection. This gives the second bound. Finally, angular error at most ϵ implies Euclidean error at most ϵ , since $2 \sin(\theta/2) \leq \theta$. Integrating the centered bound over an independent output law proves the last assertion. $\square$

For probability measures P, Q , relative entropy means

$$D(P\|Q) = \begin{cases} \int \log \frac{dP}{dQ} dP, & P \ll Q, \\ +\infty, & \text{otherwise.} \end{cases}$$

For a countable discrete variable W , $H(W)$ is its Shannon entropy in nats. Mutual information and conditional mutual information are the corresponding relative entropies of a joint law from the product of its marginals, conditionally when appropriate. We use regular conditional laws on standard Borel spaces. The conditional chain rule and average conditional relative-entropy formulas used here are given by Austin [1, Lemma 3.6 and Equations (13)–(18)]. For probability data processing under a measurable map or a common Markov kernel, see van Erven and Harremoës [11, Theorems 5 and 9; Example 2]. Our finite-reference comparisons use the following precise normalization.

Lemma 2.2 (A finite reference measure). *Let P be a probability measure and let Q be a finite nonzero measure on the same measurable space. Put $c = Q(\Omega)$ and $\bar{Q} = Q/c$. When $P \ll Q$, define*

$$D_{\text{fin}}(P\|Q) = \int \log \frac{dP}{dQ} dP.$$

This extended quantity is bounded below by $-\log c$ and satisfies

$$D_{\text{fin}}(P\|Q) = D(P\|\bar{Q}) - \log c.$$

If T is measurable, then

$$D_{\text{fin}}(T_{\#}P\|T_{\#}Q) \leq D_{\text{fin}}(P\|Q).$$

Proof. The density identity $dP/dQ = c^{-1}dP/d\bar{Q}$ gives the displayed equality and lower bound. Both Q and $T_{\#}Q$ have mass c , and $T_{\#}\bar{Q} = T_{\#}Q$. Probability data processing applied to $(P, \bar{Q})$, followed by subtracting the same $\log c$ from both sides, proves the claim. For completeness, probability data processing follows from the identity

$$\frac{dT_{\#}P}{dT_{\#}\bar{Q}}(T) = \mathbb{E}_{\bar{Q}} \left[\frac{dP}{dQ} \middle| T \right]$$

and conditional Jensen for $x \log x$. Thus the finite comparison uses a normalized probability reference before the inequality is applied. $\square$

We will also use the entropy variational inequality. If $P \ll Q$ are probabilities and f is bounded and measurable, then

$$\mathbb{E}_P f \leq D(P\|Q) + \log \mathbb{E}_Q e^f. \quad (2)$$

Indeed, normalize $e^f Q$ to a probability measure and expand the nonnegative relative entropy from P to that measure. Truncation gives the same statement for nonnegative f whenever $\mathbb{E}_Q e^f < \infty$, and for integrable f with a finite exponential moment. This is the standard variational relation between entropy and exponential moments; see Dupuis and Mao [3, Equations (1.1)–(1.2), pp. 1–2].

2.1 Density versions for exact labels

A Gaussian block maps a spherical signal into Euclidean label space. The next lemma specifies a common measurable density version and the integrability that permits subsequent entropy subtractions. Its first-moment argument is also a separate route from cube averages to the exact density.

Lemma 2.3 (Projection densities and logarithmic integrability). *Let $1 \leq \ell < n$. If ρ is a finite measure with $\rho \ll \sigma$, then the image $g_{\#}\rho$ of ρ under every full-row-rank $g \in \mathbb{R}^{\ell \times d}$ has a Lebesgue density. For $\delta > 0$, put*

$$p_{\rho,\delta}(g, y) = (2\delta)^{-\ell} \int \mathbf{1}_{\{\|gu-y\|_{\infty} \leq \delta\}} d\rho(u).$$

A jointly Borel density version is

$$p_{\rho}(g, y) = \lim_{j \rightarrow \infty} p_{\rho, 1/j}(g, y) \quad (3)$$

where the limit exists and is finite, and zero otherwise.

Suppose now that ν is a probability measure with $\nu \leq H\sigma$, $H < \infty$, and $G \sim \gamma_{\ell}$ is independent of $S \sim \nu$. Then $\log p_{\nu}(G, GS)$ is integrable. In addition,

$$\sup_{\delta > 0} \mathbb{E} p_{\nu,\delta}(G, GS) \leq (2\pi)^{-\ell/2} \iint \|u - s\|^{-\ell} d\nu(u) d\nu(s) < \infty, \quad (4)$$

and the same upper bound holds for $\mathbb{E} p_{\nu}(G, GS)$.

Finally, let a countable W with $H(W) < \infty$ have any joint law with (S, G) that leaves the latter's product marginal unchanged. For each positive-probability w , put $\nu_w = P_{S|W=w}$. The actual conditional law of GS given (G, W) has a Lebesgue density. The logarithm of that density and $\log p_{\nu_w}(G, GS)$ are both integrable under the actual law.

Proof. The first ℓ coordinates of a uniform spherical point have density

$$\frac{\Gamma(d/2)}{\pi^{\ell/2} \Gamma((d-\ell)/2)} (1 - \|y\|^2)^{(d-\ell-2)/2} \mathbf{1}_{\{\|y\| < 1\}}.$$

To see the normalization, write the point as a standard Gaussian vector divided by its length. Its projected squared length has the Beta($\ell/2, (d-\ell)/2$) law and its projected direction is uniform. Polar coordinates give the displayed density. An orthogonal change of coordinates followed by the invertible map $(gg^{\top})^{1/2}$ gives the density for $g_{\#}\sigma$, with Jacobian $\det(gg^{\top})^{-1/2}$. Absolute continuity gives a density for $g_{\#}\rho$. Each cube average is Borel in (g, y) by integration of a Borel indicator. Lebesgue differentiation shows that (3) is a density version for each full-rank g ; the specified limit is jointly Borel.

We first verify logarithmic integrability from this explicit spherical density. Under the product law $\gamma_{\ell} \otimes \sigma$, the logarithm of the residual squared radius is integrable by the beta law. Gram–Schmidt expresses $\det(GG^{\top})$ as a product of chi-square variables with positive degrees of freedom, whose logarithms have finite absolute means. Hence $\log p_{\sigma}(G, GS)$ is integrable. Domination by H preserves this integrability under $\gamma_{\ell} \otimes \nu$. For every full-rank g , data processing gives

$$D(g_{\#}\nu \| g_{\#}\sigma) \leq D(\nu \| \sigma) \leq \log H.$$

The negative logarithmic part of any probability density ratio a/b , integrated under a , is at most $1/e$, because $t \log^{-}(t) \leq 1/e$. Its positive part is therefore finite when the relative entropy is finite. Apply this to p_{ν}/p_{σ} and add the integrable baseline logarithm.

Here is a direct proof that retains the first density moment in (4). For distinct u, s , the ℓ coordinates of $G(u - s)$ are independent centered normals with variance $\|u - s\|^2$. Their joint density is at most $(2\pi)^{-\ell/2} \|u - s\|^{-\ell}$. The cube volume is $(2\delta)^\ell$. Multiplying these quantities and applying Tonelli proves the first inequality in (4); the diagonal is null since $\nu \leq H\sigma$ is nonatomic. For each $s \in S^{d-1}$, Lemma 2.1 and dyadic shells give

$$\int \|u - s\|^{-\ell} d\nu(u) \leq 1 + H2^\ell \sum_{j \geq 0} 2^{-j(n-\ell)} < \infty.$$

The term 1 covers distances greater than one, and $n - \ell > 0$ makes the series converge. This is uniform in s .

For each full-rank G , Lebesgue differentiation holds at GS for ν -almost every S , since $G_\# \nu$ has a density. Fubini and Fatou pass the first-moment bound to $p_\nu(G, GS)$. It follows that $\mathbb{E} \log^+ p_\nu(G, GS) < \infty$. To control the other part, let ϕ_ℓ be the standard Gaussian density on $\mathbb{R}^\ell$. For each G ,

$$\int p_\nu(G, y) \left(\log \frac{p_\nu(G, y)}{\phi_\ell(y)} \right)_- dy \leq \frac{1}{e}.$$

Also $\mathbb{E} \|GS\|^2 = \ell$, so $\mathbb{E}[-\log \phi_\ell(GS)] = \frac{\ell}{2} \log(2\pi) + \frac{\ell}{2}$. The inequality

$$(-\log p_\nu)_+ \leq (\log(p_\nu/\phi_\ell))_- + (-\log \phi_\ell)_+$$

proves the negative logarithmic bound directly. The densities in these conclusions describe the exact labels; the cube averages only prove their estimates.

For the last assertion, conditioning on $W = w$ dominates the joint law of (G, S) by $P(W = w)^{-1}$ times $\gamma_\ell \otimes \nu$. This gives an actual conditional image density for every full-rank G , up to the usual null set in G . The product marginal is unchanged after averaging over w , so the actual expectation of $|\log p_\nu(G, GS)|$ is finite. The relative entropy of the actual conditional image density from $p_\nu(G, \cdot)$, averaged over (G, W) , equals $I(W; GS | G) \leq H(W)$. The density-ratio negative part is at most $1/e$, hence its logarithm is integrable and so is the actual log density.

Similarly, conditional data processing gives

$$\mathbb{E} D(P_{GS|G,W} \| G_\# \nu_W) \leq \mathbb{E} D(P_{S|G,W} \| \nu_W) = I(S; G | W) \leq H(W).$$

The last inequality follows from $I(S; G) = 0$ and the chain rule. The log ratio in this display is integrable by the same negative-part bound. Subtracting it from the actual log density proves integrability of $\log p_{\nu_W}(G, GS)$. The argument also applies to a deterministic refinement $W' = (W, E(S, W))$ whenever $H(W') < \infty$. $\square$

2.2 A message selected using actual rows

Let $m, r \geq 1$, $\ell = m + r < n$, and let $S \sim \sigma$. Let $A \sim \gamma_m$ be independent of S , and let W be a countable message with $H(W) < \infty$, coupled arbitrarily to (S, A) . We define two laws:

- In the *aligned experiment* P_1 , retain the joint law of (S, A, W) , take $C \sim \gamma_r$ independently, and set $G = (A; C)$ and $Y = GS$.
- In the *independent experiment* P_0 , retain the marginal law of (S, W) , draw $G \sim \gamma_\ell$ independently of that pair, and again set $Y = GS$.

Both laws have the same (S, W) marginal and the same product (S, G) marginal. In the aligned law, W may be dependent on the rows G ; in the independent law, G is independent of (S, W) . Define

$$J_i(W) = I_i(S; W \mid G, Y), \quad i = 0, 1,$$

and, for positive-probability w , let $\nu_w = P_{S|W=w}$, which is common to both experiments.

Proposition 2.4 (The information identity and its density bound). *In the two experiments just defined, all quantities below are finite, and*

$$J_0(W) - J_1(W) = h_0(Y \mid G, W) - h_1(Y \mid G, W) - I_1(S; G \mid W) \quad (5)$$

$$\leq h_0(Y \mid G, W) + \mathbb{E}_1 \log p_{\nu_W}(G, Y). \quad (6)$$

If $E = E(S, W)$ is countable and $H(E \mid W) < \infty$, and the same deterministic refinement $W' = (W, E)$ is used in both laws, then

$$J_0(W) - J_1(W) \leq J_0(W') - J_1(W') + H(E \mid W). \quad (7)$$

Proof. Because Y is determined by (S, G) , the conditional chain rule gives

$$J_i(W) = I_i(S; W \mid G) - I_i(W; Y \mid G).$$

Since S and G are independent in both laws,

$$I_i(S; W \mid G) = I(S; W) + I_i(S; G \mid W),$$

and $I_0(S; G \mid W) = 0$. The (S, W) and (S, G) marginals agree, so subtraction gives

$$J_0(W) - J_1(W) = I_1(W; Y \mid G) - I_0(W; Y \mid G) - I_1(S; G \mid W).$$

All mutual information terms are finite: those involving the countable message are at most $H(W)$, and the last is at most $H(W)$ by this same chain rule. Lemma 2.3 makes the conditional differential entropies finite. Expressing the two $I(W; Y \mid G)$ terms as entropy differences, whose $h(Y \mid G)$ terms agree, proves (5). In particular the conditional row-information term has a negative sign.

Apply conditional data processing to $s \mapsto Gs$:

$$\begin{aligned} -h_1(Y \mid G, W) - \mathbb{E}_1 \log p_{\nu_W}(G, Y) &= \mathbb{E}_1 D(P_1(Y \in \cdot \mid G, W) \| G_{\#} \nu_W) \\ &\leq \mathbb{E}_1 D(P_1(S \in \cdot \mid G, W) \| \nu_W) \\ &= I_1(S; G \mid W). \end{aligned}$$

The integrability assertion of the same lemma justifies these subtractions. Substituting this inequality into (5) proves (6).

Finally

$$J_i(W') = J_i(W) + I_i(S; E \mid W, G, Y).$$

The difference for W therefore equals the difference for W' minus the $i = 0$ term and plus the $i = 1$ term. The former is nonnegative and the latter is at most $H(E \mid W)$. This proves (7). $\square$

3 A projection moment with shared and fresh rows

The information bound in Proposition 2.4 contains a projection density evaluated at the true label. The rows used to select the message can be biased after conditioning on that message. We first bound a high moment under the original Gaussian row law. Some rows are shared across all factors in this moment; the other rows are averaged independently in each factor. The high moment will then pay for the change from Gaussian rows to selected rows.

Expanding a projection moment into independent signal copies is part of the projection method of Sharan, Sidford, and Valiant [9, Section 7, Lemma 12]. The estimate below treats exact multirow projections. Its geometric hypothesis has two local mass bounds: one controls distance to a span, and the other controls the small radial scales.

For $x > 0$, write $\log^+ x = \max\{\log x, 0\}$.

Theorem 3.1 (Mixed projection moment and exact density version). *Let $n = d - 1$, let $m, r, q \geq 1$ be integers, and put $\ell = m + r$. Suppose*

$$\ell < n, \quad \ell - m - (q - 1) \geq 1. \quad (8)$$

Let ρ be a finite nonnegative Borel measure on S^{d-1} . Suppose that $B, D > 0$ and, for every $z \in \mathbb{R}^d$ and $t > 0$,

$$\rho(B(z, t)) \leq Bt^\ell, \quad \rho(B(z, t)) \leq Dt^n. \quad (9)$$

Let $X \sim \gamma_m$ and $Z \sim \gamma_r$ be independent, and set $G = (X; Z)$. For $\delta > 0$, define the cube average

$$p_{\rho, \delta}(g, y) = (2\delta)^{-\ell} \int \mathbf{1}_{\{\|gu-y\|_\infty \leq \delta\}} d\rho(u).$$

There is an absolute C such that, for every fixed $s \in S^{d-1}$ and every $\delta > 0$,

$$[\mathbb{E}_X (\mathbb{E}_Z p_{\rho, \delta}((X; Z), (X; Z)s))^q]^{1/q} \leq e^{Cd} B(1 + \log^+(D/B)). \quad (10)$$

If in addition $\rho \ll \sigma$, use the jointly Borel density p_ρ in (3). There is a measurable set $E_\rho \subset S^{d-1}$, with $\rho(E_\rho) = 0$, such that for every $s \notin E_\rho$ the cube averages converge to $p_\rho(G, Gs)$ for γ_ℓ -almost every G , and

$$[\mathbb{E}_X (\mathbb{E}_Z p_\rho((X; Z), (X; Z)s))^q]^{1/q} \leq e^{Cd} B(1 + \log^+(D/B)). \quad (11)$$

When $\rho \neq 0$, E_ρ may also be chosen so that $0 < p_\rho(G, Gs) < \infty$ for γ_ℓ -almost every G and every $s \notin E_\rho$.

Proof. We begin with the inverse-distance integral that appears after expanding the moment. Fix $s \in S^{d-1}$ and a linear subspace $V \subset \mathbb{R}^d$ of dimension $h \leq q - 1$. On a radial shell

$$R/2 < \|u - s\| \leq R, \quad R = 2^{1-i}, \quad i \geq 0,$$

consider the points satisfying $\text{dist}(u - s, V) \leq \eta R$, where $0 < \eta \leq 1$. Project them onto the radius- R ball in $s + V$. An ηR -net of this ball has at most $(3/\eta)^h$ points: a maximal separated set has disjoint h -balls of radius $\eta R/2$, all lying in the ball of radius $R + \eta R/2$. For $h = 0$, take the single point of the zero-dimensional ball. Every point under consideration is within $2\eta R$ of one of the net points. The ambient-center bounds in (9) therefore give

$$\rho\{R/2 < \|u - s\| \leq R, \text{ dist}(u - s, V) \leq \eta R\} \leq e^{Cd} \min\{BR^\ell \eta^{\ell-h}, DR^n \eta^{n-h}\}. \quad (12)$$

Both powers of η are positive, so the part of the shell at zero distance from V has zero ρ -mass. The ball bounds also give $\rho(\{s\}) = 0$, by sending their radius to zero.

On the band $2^{-a-1}R < \text{dist}(u - s, V) \leq 2^{-a}R$, $a \geq 0$, we have

$$\|u - s\|^{-r} \text{dist}(u - s, V)^{-m} \leq 2^{r+m} R^{-\ell} 2^{am}.$$

Summing (12) over these bands with either of its two terms gives a geometric series. The relevant exponents satisfy

$$\ell - h - m \geq \ell - (q - 1) - m \geq 1, \quad n - h - m \geq 1.$$

Thus the integral on one radial shell is at most

$$e^{Cd} \min\{B, DR^{n-\ell}\}.$$

The shells $R = 2^{1-i}$ cover all nonzero distances on the unit sphere. Put $b = n - \ell \geq 1$. Their sum obeys

$$\sum_{i \geq 0} \min\{B, D2^{b(1-i)}\} \leq CB(1 + \log^+(D/B)).$$

For example, after the $i = 0$ term, if $D/B \leq 1$ the terms form a geometric tail of ratio at most $1/2$. If $D/B > 1$, at most $1 + \log_2(D/B)/b$ further terms equal B , followed by such a tail. We have proved the uniform estimate

$$\int \|u - s\|^{-r} \text{dist}(u - s, V)^{-m} d\rho(u) \leq e^{Cd} B(1 + \log^+(D/B)). \quad (13)$$

We now expand the projection moment. For fixed $u \neq s$, the r coordinates of $Z(u - s)$ are independent centered normals of variance $\|u - s\|^2$. Bounding each cube probability by its volume times the peak density shows that the fresh-row average, after division by $(2\delta)^r$, is at most $(2\pi)^{-r/2} \|u - s\|^{-r}$. Expand the q th power of the remaining integral over $u_1, \dots, u_q$. Write $v_i = u_i - s$. For one shared row of X , the vector of its q evaluations on the v_i is Gaussian with covariance $\text{Gram}(v_1, \dots, v_q)$. On the set where this matrix is nonsingular, its peak density bounds its cube probability by

$$(2\delta)^q (2\pi)^{-q/2} \det \text{Gram}(v_1, \dots, v_q)^{-1/2}.$$

The singular tuples have zero $\rho^{\otimes q}$ -mass: successively, (12) gives zero mass to $s + \text{span}(v_1, \dots, v_{i-1})$. Independence of the m shared rows and the r fresh rows cancels all cube volumes and bounds the expanded moment by

$$(2\pi)^{-\ell q/2} \int \prod_{i=1}^q \|v_i\|^{-r} \det \text{Gram}(v_1, \dots, v_q)^{-m/2} d\rho^{\otimes q}. \quad (14)$$

Gram-Schmidt gives

$$\det \text{Gram}(v_1, \dots, v_q) = \prod_{i=1}^q \text{dist}(v_i, \text{span}(v_1, \dots, v_{i-1}))^2,$$

with the empty span equal to $\{0\}$. Integrate in the reverse order $u_q, \dots, u_1$. Every inner integral is bounded by (13), uniformly in the previous vectors. This proves (10). All integrands used in the expansion are nonnegative, so Tonelli justifies each interchange.

Suppose now that $\rho \ll \sigma$. For every full-rank g , Lemma 2.3 and Lebesgue differentiation give cube convergence outside a Lebesgue-null subset of label space. Its inverse image is ρ -null because

$g_{\#}\rho$ has a density. The same argument shows that the exact density is finite and, when $\rho \neq 0$, positive at gs for ρ -almost every s . Since γ_ℓ gives full row rank almost surely, Fubini supplies a single ρ -null set E_ρ with these properties for every $s \notin E_\rho$ and almost every independent G .

Fix such an s . Fatou in the fresh rows gives

$$\mathbb{E}_Z p_\rho((X; Z), (X; Z)s) \leq \liminf_{j \rightarrow \infty} \mathbb{E}_Z p_{\rho, 1/j}((X; Z), (X; Z)s)$$

for almost every X . Raise to the q th power and use Fatou again in X . The uniform bound (10) then proves (11). This last assertion has the stated ρ -almost-everywhere qualification. $\square$

3.1 Changing the actual row law before taking the density limit

The exceptional set in the exact moment theorem is measured using ρ . A selected message produces a different conditional law of the true signal and the actual rows. The following form uses the uniform cube estimate for every signal, pays for that row selection, and then takes the density limit under the actual label law. The localization application uses this order in its own specialized proof.

Proposition 3.2 (Actual-row tilt and exact-label limit). *Keep the dimensions of Theorem 3.1. Let $\nu \leq H\sigma$, $H < \infty$, be a probability measure. Let $S \sim \nu$ and $X \sim \gamma_m$ be independent, let W be countable with $H(W) < \infty$ and any joint law with (S, X) , and let $Z \sim \gamma_r$ be independent of (S, X, W) . Put $G = (X; Z)$ and $Y = GS$. For every positive-probability w , let $\rho_w \ll \sigma$ be a finite nonnegative measure satisfying (9) with constants $B_w, D_w > 0$, and set*

$$C_w = e^{Cd} B_w (1 + \log^+(D_w/B_w)),$$

where C is the constant in the theorem. Assume $\mathbb{E} \log(1 + C_W) < \infty$.

The actual conditional law of Y given (G, W) has a Lebesgue density. With the versions (3),

$$p_{\rho_w, 1/j}(G, Y) \longrightarrow p_{\rho_w}(G, Y) \quad \text{almost surely under the actual law.}$$

For every $\delta > 0$, and also for the exact density,

$$\mathbb{E} \log(1 + p_{\rho_w, \delta}(G, Y)) \leq \frac{I(X; W | S)}{q} + \mathbb{E} \log(1 + C_w), \quad (15)$$

$$\mathbb{E} \log(1 + p_{\rho_w}(G, Y)) \leq \frac{I(X; W | S)}{q} + \mathbb{E} \log(1 + C_w) \leq \frac{H(W)}{q} + \mathbb{E} \log(1 + C_w). \quad (16)$$

In particular $\log^+ p_{\rho_w}(G, Y)$ is integrable, and the extended expectation of $\log p_{\rho_w}(G, Y)$, with $\log 0 = -\infty$, is defined and bounded above by the same right-hand side. For a deterministic countable refinement $W' = (W, E(S, W))$ of finite entropy, the row charge is unchanged:

$$I(X; W' | S) = I(X; W | S).$$

Proof. For almost every (s, w) , let $K_{s, w}$ be the conditional law of X . Since X and S are initially independent, the conditional chain rule gives

$$\mathbb{E}_{S, W} D(K_{S, W} \| \gamma_m) = I(X; W | S) \leq H(W) < \infty. \quad (17)$$

Thus $K_{s, w} \ll \gamma_m$ for almost every (s, w) . For fixed s, w, δ , define

$$F_{w, \delta}(s, x) = \mathbb{E}_Z p_{\rho_w, \delta}((x; Z), (x; Z)s).$$

The every-signal estimate (10) says that $\|F_{w,\delta}(s, \cdot)\|_{L^q(\gamma_m)} \leq C_w$, uniformly in δ . Conditional on (S, X, W) , Z still has its independent Gaussian law. Jensen in Z , followed by (2) with the nonnegative test $q \log(1 + F_{w,\delta})$, gives

$$\begin{aligned} \mathbb{E}[\log(1 + p_{\rho_{w,\delta}}(G, Y)) | S = s, W = w] &\leq \mathbb{E}_{K_{s,w}} \log(1 + F_{w,\delta}(s, X)) \\ &\leq \frac{D(K_{s,w} \| \gamma_m)}{q} + \frac{1}{q} \log \mathbb{E}_{\gamma_m}(1 + F_{w,\delta}(s, X))^q \\ &\leq \frac{D(K_{s,w} \| \gamma_m)}{q} + \log(1 + C_w). \end{aligned}$$

The last step is the L^q triangle inequality, valid since $q \geq 1$. Average over (S, W) and use (17) to obtain (15).

It remains to specify the null set for the limit in the actual experiment. The marginal law of (G, S) is $\gamma_\ell \otimes \nu$. Lemma 2.3 therefore gives a Lebesgue density for the actual conditional law of Y given (G, W) . For each w and every full-rank g , Lebesgue differentiation for $g_{\#} \rho_w$ says that $p_{\rho_{w,1/j}}(g, y) \rightarrow p_{\rho_w}(g, y)$ outside a Lebesgue-null set of y . The actual conditional density gives that set zero probability. Gaussian G has full rank almost surely, and there are only countably many w , proving the asserted almost-sure convergence under the actual law.

The functions $\log(1 + p_{\rho_{w,1/j}}(G, Y))$ are nonnegative. Fatou applied now, after the row-law change, passes (15) to (16). Since $\log^+ x \leq \log(1 + x)$, the positive logarithmic part is integrable; the negative part may have infinite mean, in which case the extended expectation of $\log p_{\rho_w}$ is $-\infty$ and the upper bound remains valid. Applications that subtract this logarithm verify its full integrability separately. Finally, E is determined by (S, W) , so $I(X; E | S, W) = 0$; the chain rule proves the refinement identity. $\square$

Remark 3.3 (The parameters used by the localization potential). The inverse-distance potential in the localization companion [5, Section 12, Lemma 12.5] takes

$$m = r = q = k = \lfloor d/16 \rfloor, \quad \ell = p = 2k, \quad n = d - 1.$$

For its bounded-density probability $\nu \leq H\sigma$, it defines $J_\nu(s) = \int \|u - s\|^{-p} d\nu(u)$ and $d\rho = d\nu/J_\nu$. The application proves locally that

$$\rho(B(z, t)) \leq 2^p t^p, \quad \rho(B(z, t)) \leq 2^{p+n} H t^n.$$

Thus the precise constants in Theorem 3.1 are $B = 2^p$ and $D = 2^{p+n} H$, the angular margin is $p - k - (k - 1) = 1$, and the radial margin is $n - p > 0$. The theorem supplies the uniform cube estimate and its exact-density scope. The localization application imports the positive-scale estimate and proves its reweighting, specialized row-law change, and actual-law Fatou passage locally, together with its potential calculations. Proposition 3.2 gives a general comparison of the same type within the present paper.

4 A critical radius makes the density terms cancel

The two terms in (6) can each be large when the message concentrates the signal on a small region. We will compare them at the same region and scale. A discrete refinement records a radius where the posterior mass is largest relative to the projection dimension, together with its logarithmic mass level. Confinement at that radius lowers the entropy in the independent experiment. The same level raises the aligned projection density by the opposite amount.

Throughout this section d is sufficiently large and

$$m = k = \lfloor d/10 \rfloor, \quad \ell = \lfloor d/2 \rfloor, \quad r = \ell - m, \quad n = d - 1. \quad (18)$$

The integer k is a moment order. We reserve q in this section for a logarithmic mass level. In particular $\ell < n$, $\ell - k - m \geq 1$, and $n - k - m \geq 1$.

Theorem 4.1 (Critical-radius comparison). *In the two experiments of Subsection 2.2, use the dimensions (18). Thus $S \sim \sigma$, $A \sim \gamma_m$ is independent of S , and W is countable with an arbitrary joint dependence on (S, A) . The aligned law retains (S, A, W) and appends independent r -row C , whereas the independent law retains (S, W) and draws an independent ℓ -row matrix G . If $H(W) \leq d^2$, then, for an absolute constant K ,*

$$I_0(S; W \mid G, GS) \leq I_1(S; W \mid G, GS) + Kd. \quad (19)$$

Proof. For a positive-probability message w , write

$$p_w = \mathbb{P}(W = w), \quad D_w = p_w^{-1}, \quad F_w = P_{S|W=w}.$$

Then $F_w \leq D_w \sigma$. The common density lemma and Proposition 2.4 apply to every finite-entropy refinement used below.

Selecting a radius and a mass level

Let $R_j = 2^{1-j}$ for $j \geq 0$, and define

$$b_w(s) = \max_{j \geq 0} R_j^{-\ell} F_w(B(s, R_j)). \quad (20)$$

The $j = 0$ term is $2^{-\ell}$. On the other hand, Lemma 2.1 gives

$$R_j^{-\ell} F_w(B(s, R_j)) \leq 2^n D_w R_j^{n-\ell} \longrightarrow 0.$$

Hence the maximum is attained. Let $j_w(s)$ be the least maximizing index, and put $q_w(s) = \lceil \log b_w(s) \rceil$. Ball masses are measurable functions of their centers, and a countable comparison selects the least maximizing index, so these choices are measurable.

The preceding upper bound and $b_w \geq 2^{-\ell}$ give

$$0 \leq j_w(s) \leq C(d + 1 + \log D_w).$$

For $R_j \leq 1$, the same upper bound is at most $2^n D_w$; for $R_j > 1$, total mass one suffices. Therefore

$$-Cd \leq q_w(s) \leq C(d + 1 + \log D_w).$$

Define the discrete refinement

$$E = (j_W(S), q_W(S)), \quad W' = (W, E).$$

For a positive-probability value w' , write $F_{w'} = P_{S|W'=w'}$. There are at most $C(d + 1 + \log D_w)^2$ possible pairs conditional on $W = w$. Since $\mathbb{E} \log D_W = H(W)$, counting those pairs and then using Jensen gives

$$H(E \mid W) \leq C \log(d + 1 + H(W)). \quad (21)$$

In particular $H(W') = H(W) + H(E | W) < \infty$. The same deterministic function of (S, W) is used in both experiments, so (7) gives

$$J_0(W) - J_1(W) \leq J_0(W') - J_1(W') + H(E | W). \quad (22)$$

Fix a positive-probability $w' = (w, j, q)$ and put

$$\eta = \mathbb{P}(E = (j, q) | W = w), \quad F = P_{S|W'=w'}.$$

The measure F is the restriction of F_w to the selected class, divided by η . It satisfies

$$F \leq D'\sigma, \quad F(B(z, t)) \leq B't^\ell \quad (z \in \mathbb{R}^d, t > 0), \quad D' = \frac{D_w}{\eta}, \quad B' = \frac{4^\ell e^q}{\eta} \geq 1. \quad (23)$$

For the local mass bound, if $t \leq 1$ and $B(z, t)$ has positive F -mass, choose a selected-class point s_0 in that ball. Choose a dyadic radius $R_h \leq 2$ satisfying $2t \leq R_h \leq 4t$; its ball about s_0 covers $B(z, t)$. Its F_w -mass is at most $e^q(4t)^\ell$ by (20). Divide by η . For $t > 1$, use total mass one and $B' \geq 1$. The latter bound follows from $e^q \geq b_w \geq 2^{-\ell}$ and $\eta \leq 1$.

The independent projection loses the mass level

The selected class can be covered by at most

$$N \leq e^{1-q} R_j^{-\ell} \quad (24)$$

balls of radius $2R_j$. Indeed, choose successive class points at mutual distances greater than $2R_j$. Their radius- R_j balls are disjoint. Each such ball has F_w -mass at least $e^{q-1} R_j^\ell$, since j maximizes the ratio and $q = \lceil \log b_w \rceil$. Total mass one bounds the number of points and forces the selection to terminate with the asserted cover. For each of the countably many labels w' , fix one such finite cover and assign a point to the first ball containing it. The resulting center label c is measurable and has conditional entropy at most $\log N$.

In the independent experiment, conditional on $W' = w'$, the matrix G is independent of (S, c) and $\|S - c\| \leq 2R_j$. Consequently

$$\mathbb{E}_0[\|G(S - c)\|^2 | W' = w'] \leq 4\ell R_j^2.$$

Compare the conditional density of $Y - Gc$, given (G, c, w') , with the centered Gaussian density of covariance $4R_j^2 I_\ell$. Nonnegative relative entropy and the displayed second moment give

$$h_0(Y | G, c, W' = w') \leq \ell \log R_j + C\ell.$$

The conditional densities and their logarithms are integrable by Lemma 2.3, also after the finite center refinement. Removing c costs at most $\log N$. Using (24) cancels the radius and gives

$$h_0(Y | G, W' = w') \leq -q + Cd. \quad (25)$$

The aligned density gains the same level

For the same F , the exact moment bound needed here is

$$\mathbb{E}_A(\mathbb{E}_{CP_F}((A; C), (A; C)s))^k \leq [C^d B'(1 + \log D')]^k \quad \text{for } F\text{-almost every } s. \quad (26)$$

To check the exact substitution in Theorem 3.1, use m shared rows, r fresh rows, moment order k , and $\ell = m + r$. The angular gap is $\ell - m - (k - 1) = \ell - m - k + 1 \geq 2$, and $\ell < n$. The first growth coefficient is $B = B'$. The second is $D = 2^n D'$, by $F \leq D' \sigma$ and Lemma 2.1. Since $B', D' \geq 1$,

$$e^{C^d B' [1 + \log^+(2^n D' / B')]} \leq C^d B' (1 + \log D').$$

The exact conclusion of the theorem yields (26).

In the aligned law, conditioning on (S, W') can bias the actual rows A . Its precise average cost is

$$\mathbb{E}_1 D(P_1(A \in \cdot \mid S, W') \parallel \gamma_m) = I_1(A; W' \mid S) = I_1(A; W \mid S) \leq H(W). \quad (27)$$

Here A is initially independent of S , and E is determined by (S, W) . The finite average divergence also gives absolute continuity of the conditional actual row law for almost every (S, W') .

For w' fixed, the actual marginal $S \mid W' = w'$ is exactly F . Hence the F -null exceptional set in (26) has zero actual probability. For such an s , set

$$K_F(A, s) = \mathbb{E}_C \log p_F((A; C), (A; C)s).$$

The reference and actual logarithmic integrability follow from Lemma 2.3; null sets in A transfer to its conditional actual law by the preceding absolute continuity. Jensen and (26) imply

$$\mathbb{E}_A e^{k K_F(A, s)} \leq [C^d B' (1 + \log D')]^k.$$

Apply (2) to $k K_F$ under the conditional actual row law relative to γ_m . The rows C remain independent after conditioning on (S, W', A) . Averaging and using (27) therefore gives

$$\mathbb{E}_1 \log p_{F_{W'}}(G, Y) \leq \frac{H(W)}{k} + Cd + \mathbb{E}q + H(E \mid W) + \log(1 + H(W')). \quad (28)$$

Indeed,

$$\log B' = \ell \log 4 + q + \log(1/\eta), \quad \mathbb{E} \log(1/\eta) = H(E \mid W), \quad \mathbb{E} \log D' = H(W'),$$

and Jensen bounds $\mathbb{E} \log(1 + \log D')$ by $\log(1 + H(W'))$. These expectations are finite by the preceding entropy bounds.

We now have the two estimates at the same selected radius and mass level. Apply (6) to W' , average (25), and use (28). The terms $-\mathbb{E}q$ and $+\mathbb{E}q$ cancel. Including the refinement charge (22) yields the more precise inequality

$$J_0(W) - J_1(W) \leq \frac{H(W)}{k} + Cd + 2H(E \mid W) + \log(1 + H(W')). \quad (29)$$

Now $k = \lfloor d/10 \rfloor$, $H(W) \leq d^2$, and (21) make the right-hand side at most Kd for an absolute K . This proves (19). $\square$

5 Posterior density levels and auxiliary projections

A state can be informative because its posterior concentrates on a small part of the sphere. The two comparisons in this section measure that concentration at dyadic radii. In each comparison, a state is tested after independent Gaussian observations have been supplied to the analysis. Replacing some of those rows by the current block makes the state update a conditional Markov channel. The task is to bound the information cost of returning from the current rows to independent ones.

The two arguments use different refinements. The first retains only the level of the largest normalized ball mass and uses a net to estimate the independent projection entropy. The second retains both the first maximizing radius and the density level; its covering estimate cancels the radius directly. These choices lead to different row counts and to complementary exact-density estimates.

We use the measure, Gaussian law γ_t , and entropy conventions of Section 2. Locally, write $N = d - 1$ for the sphere dimension; this avoids confusing it with the actual-row count n in the second comparison. The row symbols have the following roles, with their numerical values specified in each subsection:

Comparison	Total rows	Actual rows	Fresh rows	Moment
One level	m	k	$j = m - k$	q
Two labels	k	n	$l = k - n$	q

Constants C, C_0 are positive and absolute and may increase. All claims are for sufficiently large d , with an absolute threshold unless otherwise stated.

We use the local spherical-ball and projection-density results, Lemmas 2.1 and 2.3. In particular, for every $z \in \mathbb{R}^d$ and $\rho > 0$,

$$\sigma(B(z, \rho)) \leq (2\rho)^N. \quad (30)$$

For $\nu \leq K\sigma$ and a full-row-rank matrix g with fewer than N rows, write $p_\nu(g, \cdot)$ for the jointly measurable cube-average version of the density of gS , $S \sim \nu$. For independent Gaussian G and $S \sim \nu$, the logarithm $\log p_\nu(G, GS)$ is integrable. The same lemma supplies the actual conditional image densities and the integrable log ratios used below when a finite state is correlated with the rows. These density statements concern the exact observations.

We use (2) with $f = q \log F$ in the form

$$\mathbb{E}_P \log F \leq \frac{D(P||Q) + \log \mathbb{E}_Q F^q}{q}, \quad P \ll Q, \quad q > 0, \quad F \geq 0. \quad (31)$$

Truncation gives the integrable form and the upper bound when the left side is negative infinite.

5.1 A comparison using one density level

Set, locally in this subsection,

$$m = \lfloor d/4 \rfloor, \quad k = \lfloor d/32 \rfloor, \quad j = m - k, \quad q = \lfloor d/16 \rfloor. \quad (32)$$

The auxiliary projection has m rows, of which k can be actual rows and j remain independent. The integer q is the moment order. For large d ,

$$m < N, \quad m - k - (q - 1) \geq 1.$$

For a probability measure $\nu \leq K\sigma$, $K < \infty$, put $r_i = 2^{1-i}$, $i \geq 0$, and define

$$L_\nu(s) = \max_{i \geq 0} r_i^{-m} \nu(B(s, r_i)). \quad (33)$$

The $i = 0$ term is 2^{-m} . On the other hand,

$$r_i^{-m} \nu(B(s, r_i)) \leq K 2^N r_i^{N-m},$$

which tends to zero uniformly in s . Hence the maximum is attained among at most $C(1 + d + \log K)$ initial scales, and

$$2^{-m} \leq L_\nu(s) \leq Ke^{Cd}.$$

The ball-mass functions are Borel, by integration of the Borel indicator $\mathbf{1}_{\{\|s-u\| \leq r_i\}}$. Thus L_ν and the first maximizing scale are Borel. For $a = 2^b$, $b \in \mathbb{Z}$, let

$$F_a = \{s : a \leq L_\nu(s) < 2a\}, \quad w_a = \nu(F_a).$$

Only $C(1 + d + \log K)$ levels can be nonempty, and a nonempty level has $a \geq 2^{-m}$.

Lemma 5.1 (Independent projection entropy from a net). *If $G \sim \gamma_m$ and $S \sim \nu \leq K\sigma$ are independent, then*

$$\mathbb{E} \log p_\nu(G, GS) \geq \mathbb{E} \log L_\nu(S) - Cd - \log(C(1 + d + \log K)). \quad (34)$$

Proof. At each relevant scale i , take a maximal r_i -separated set of centers on the sphere, and assign each point to a nearest center, breaking ties in a fixed order. Denote the cells and centers by C_{ih} and c_{ih} . Each cell is contained in $B(c_{ih}, r_i)$. At most 7^d centers lie within $3r_i$ of a fixed center: the ambient balls of radius $r_i/2$ about those centers have disjoint interiors and lie in a ball of radius $7r_i/2$.

Let b_{ih} be the sum of the ν -masses of cells with centers within $3r_i$ of c_{ih} . For $s \in C_{ih}$, $\nu(B(s, r_i)) \leq b_{ih}$, and $\sum_h b_{ih} \leq 7^d$. Partition the sphere by its first maximizing scale $i(s)$ and its cell at that scale. Write w_{ih} for the masses of these classes, omitting zero-mass classes. If $N_0 \leq C(1 + d + \log K)$ is the number of relevant scales, Jensen gives

$$\mathbb{E} \log \frac{\nu(B(S, r_{i(S)}))}{w_{i(S), h(S)}} \leq \log \sum_{i, h: w_{ih} > 0} b_{ih} \leq d \log 7 + \log N_0. \quad (35)$$

In a class (i, h) , let ν_{ih} be the conditional law of S . Its points are within r_i of c_{ih} . Comparing the density of GS with the Gaussian density of mean Gc_{ih} and covariance $r_i^2 I_m$, nonnegative relative entropy yields

$$\mathbb{E}_{G, S \sim \nu_{ih}} \log p_{\nu_{ih}}(G, GS) \geq m \log(1/r_i) - Cm,$$

because independence gives $\mathbb{E} \|G(S - c_{ih})\|^2 \leq mr_i^2$. Also $p_\nu \geq w_{ih} p_{\nu_{ih}}$ almost everywhere. Average its logarithm under each class. The identity $L_\nu(S) = r_{i(S)}^{-m} \nu(B(S, r_{i(S)}))$ then gives

$$\mathbb{E} \log p_\nu(G, GS) \geq \mathbb{E} \log L_\nu(S) - \mathbb{E} \log \frac{\nu(B(S, r_{i(S)}))}{w_{i(S), h(S)}} - Cm.$$

Apply (35). The refinement is finite, and every class is dominated by a finite multiple of σ ; Lemma 2.3 justifies all logarithmic expectations. $\square$

The next bound uses the same density level to control the projection when the k rows will later be allowed to depend on the state. The order of averaging is essential: the j independent rows are averaged before taking a q -th moment in the other rows.

Lemma 5.2 (A moment on one level). *Let $w_a > 0$ and $\mu = \nu(\cdot \mid F_a)$. For independent $X \sim \gamma_k$, $B \sim \gamma_j$, and $G = (X; B)$,*

$$\left[\mathbb{E}_X (\mathbb{E}_B p_\mu(G, GS))^q \right]^{1/q} \leq e^{Cd} \frac{a}{w_a} (1 + d + \log K) \quad (36)$$

for μ -almost every s .

Proof. For every ambient ball and $t > 0$,

$$\mu(B(z, t)) \leq C_0^d \frac{a}{w_a} t^m, \quad \mu(B(z, t)) \leq C_0^d \frac{K}{w_a} t^N \quad (37)$$

with one absolute C_0 . For the first estimate, if the ball has positive μ -mass, choose $s_0 \in F_a \cap B(z, t)$. For $t \leq 1$, a dyadic radius between $2t$ and $4t$ contains $B(z, t) \cap F_a$, and its ν -mass is at most $2a(4t)^m$. For $t > 1$, use total mass one and $a \geq 2^{-m}$. The second estimate follows from $\mu \leq (K/w_a)\sigma$ and (30).

Apply the shared/fresh moment theorem, Theorem 3.1, with k shared rows, j fresh rows, total row count m , moment order q , and

$$B_0 = C_0^d a/w_a, \quad D_0 = C_0^d K/w_a.$$

Its angular gap is $m - k - (q - 1) \geq 1$, and its radial gap is $N - m > 0$. The inverse factor in its one-point integration is exactly

$$\|u - s\|^{-j} \text{dist}(u - s, V)^{-k}, \quad \dim V \leq q - 1.$$

The ratio $D_0/B_0 = K/a$ does not contain w_a , and $\log^+(K/a) \leq \log K + m \log 2$. The theorem therefore gives the right side of (36) uniformly in the cube-average scale, for every fixed s .

For completeness, the passage to the exact density has the almost-everywhere scope asserted here. At G s the cube averages converge for almost every (G, s) under $\gamma_m \otimes \mu$: condition on G , use absolute continuity of its image law and Lebesgue differentiation, and then apply Fubini. For μ -almost every fixed s , Fatou first in B and then in X after taking the q -th power gives (36). This passage estimates the exact label; it does not change the learner's observation. The expansion into independent points underlying the moment theorem is related to the projection-moment method of Sharan, Sidford, and Valiant [9, Section 7, Lemma 12]; the multirow estimate used here is the local theorem with the displayed parameters. $\square$

We now specify both experiments. For a finite variable V jointly distributed with $S \sim \sigma$, define

$$\mathcal{D}(V) = I_{\text{ind}}(S; V \mid G, GS), \quad G \sim \gamma_m \text{ independent of } (S, V).$$

In the actual experiment, $X \sim \gamma_k$ is independent of $S \sim \sigma$, U is any finite variable jointly distributed with (S, X) , and $B \sim \gamma_j$ is independent of (S, X, U) . Put $G = (X; B)$ and

$$\tilde{\mathcal{D}}(U) = I_{\text{act}}(S; U \mid G, GS).$$

The same definition applies to a finite refinement V determined by (S, U) . Thus G and S are marginally independent in both experiments, but G can depend on U in the actual one. The independent experiment uses the same (S, U) marginal as the actual one.

Proposition 5.3 (One-level auxiliary-row comparison). *For the preceding experiments and parameters,*

$$\mathcal{D}(U) \leq \tilde{\mathcal{D}}(U) + C \left(d + \log(1 + d + H(U)) + \frac{H(U)}{d} \right). \quad (38)$$

The proposition applies to every finite U with the specified joint law.

Proof. First take $V = U$, or a finite refinement of U determined by (S, U) . Let P be the actual law and Q the law with the same (S, V) marginal and an independent $G \sim \gamma_m$. Their (G, S) marginals also agree. Write $\nu_v = P_{S|V=v} = Q_{S|V=v}$. Proposition 2.4 gives the exact identity

$$\mathcal{D}(V) - \tilde{\mathcal{D}}(V) = I_P(V; GS | G) - I_Q(V; GS | G) - I_P(S; G | V). \quad (39)$$

Its density bound specializes to

$$\mathcal{D}(V) - \tilde{\mathcal{D}}(V) \leq \mathbb{E}_P \log p_{\nu_V}(G, GS) - \mathbb{E}_Q \log p_{\nu_V}(G, GS). \quad (40)$$

The negative term in (39) is what pays for the changed conditional image density. All quantities are finite by Lemma 2.3 and $I_P(S; G | V) \leq H(V)$.

For u of positive probability let $K_u = P(U = u)^{-1}$, so $\nu_u \leq K_u \sigma$. Let J be the integer with $S \in F_{2^J}$ for the levels of L_{ν_U} , and put $V = (U, J)$. The level count and Jensen give

$$H(J | U) \leq \mathbb{E} \log(C(1 + d + \log K_U)) \leq \log(C(1 + d + H(U))). \quad (41)$$

Under Q , refinement increases the mean log projection density by $I_Q(J; GS | G, U) \geq 0$. Applying Lemma 5.1 to each ν_u therefore yields

$$\mathbb{E}_Q \log p_{\nu_V}(G, GS) \geq \mathbb{E} \log L_{\nu_U}(S) - Cd - \log(C(1 + d + H(U))). \quad (42)$$

For the actual upper bound, condition on $S = s, V = v = (u, b)$ and write $a = 2^b$. Conditional on (s, v, X) , B is still independent with law γ_j . Jensen in B bounds the conditional mean of $\log p_{\nu_v}(G, GS)$ by the mean, under $P_{X|s,v}$, of $\log \mathbb{E}_{B p_{\nu_v}}((X; B), (X; B)s)$. Apply (31) with reference γ_k , order q , and Lemma 5.2. The conditional divergences average to

$$\mathbb{E} D(P_{X|S,V} || \gamma_k) = I_P(X; V | S) = I_P(X; U | S) \leq H(U). \quad (43)$$

The first equality uses marginal independence of X, S ; the second uses that J is determined by (S, U) . Thus the refinement creates no additional conditional row information. Finite average divergence also gives $P_{X|S,V} \ll \gamma_k$ almost everywhere. The exact moment is needed only for ν_v -almost every s , which is precisely the conditional signal law at $V = v$.

If $w_a = \nu_u(F_a)$, the logarithm of the moment bound is

$$Cd + \log a + \log(1/w_a) + \log(1 + d + \log K_u).$$

Averaging, using $\mathbb{E} \log(1/w_a) = H(J | U)$, and applying Jensen once more, we obtain

$$\mathbb{E}_P \log p_{\nu_V}(G, GS) \leq Cd + \mathbb{E} \log a + H(J | U) + \log(1 + d + H(U)) + \frac{H(U)}{q}. \quad (44)$$

The log expectations and the truncated use of (31) are justified by the integrability already established.

Finally, $\log a \leq \log L_{\nu_U}(S)$, so the local-density terms in (42) and (44) cancel. Refinement gives

$$\mathcal{D}(U) \leq \mathcal{D}(V), \quad \tilde{\mathcal{D}}(V) \leq \tilde{\mathcal{D}}(U) + H(J | U).$$

Combine these inequalities with (40) and (41). Since q is a fixed positive fraction of d for large d , the result is (38). $\square$

5.2 A comparison retaining the radius and the level

We now use more auxiliary rows and retain the first maximizing radius as a second label. This lets a covering argument estimate the independent entropy without the net-weight average above. In this subsection the local parameters are

$$k = \lfloor d/2 \rfloor, \quad n = \lfloor d/8 \rfloor, \quad l = k - n, \quad q = n. \quad (45)$$

Here n is the number of actual rows; the surface exponent remains $N = d - 1$. In particular

$$N - k \geq d/3, \quad k - q - n \geq 1, \quad N - q - n \geq 1$$

for large d .

For a finite variable V with $S \sim \sigma$, let

$$\mathcal{J}(V) = I_{\text{ind}}(S; V \mid G, GS), \quad G \sim \gamma_k \text{ independent of } (S, V).$$

In the actual experiment $A \sim \gamma_n$ is independent of $S \sim \sigma$, V has any joint law with (S, A) , and $B \sim \gamma_l$ is independent of (S, A, V) . Put $D = (A; B)$. Thus D is marginally independent of S with law γ_k , although its conditional law given (S, V) need not be Gaussian. The independent experiment defining $\mathcal{J}(V)$ preserves the actual (S, V) marginal.

Proposition 5.4 (Two-label auxiliary-row comparison). *If V is finite and $H(V) \leq d^2$ in the preceding actual experiment, then*

$$\mathcal{J}(V) \leq I_{\text{act}}(S; V \mid D, DS) + Cd. \quad (46)$$

Proof. Let P denote the actual law. We first construct the refinement and obtain the two geometric estimates it supplies. For $\pi_v = P(V = v) > 0$, write $\nu_v = P_{S|V=v} \leq \pi_v^{-1} \sigma$, and define

$$\lambda_v(s) = \max_{i \geq 0} r_i^{-k} \nu_v(B(s, r_i)), \quad r_i = 2^{1-i}.$$

Let j be the first maximizing index and let $b = \lfloor \log_2 \lambda_v(s) \rfloor$. The maximum is attained: its $i = 0$ term is 2^{-k} , while $\pi_v^{-1} 2^N r_i^{N-k}$ bounds the i -th term and tends to zero. The ball-mass functions are Borel, so both labels are measurable. The same bounds give

$$b \geq -k, \quad 2^b \leq \pi_v^{-1} C^d, \quad 2^{-k} \leq \pi_v^{-1} C^d r_j^{N-k}.$$

Because $N - k \geq d/3$, the last inequality implies $j \leq C(1 + \log(1/\pi_v)/d)$. At fixed v , the number of pairs (j, b) is at most $C(d + 1 + \log(1/\pi_v))^2$. For $W = (V, j, b)$ evaluated at $s = S$, this proves finiteness and

$$H(W \mid V) \leq C + 2 \log(d + 1 + H(V)). \quad (47)$$

For $w = (v, j, b)$ with $\pi_w = P(W = w) > 0$, set $R = r_j$. Let E_w be the corresponding subset of the sphere, and write

$$\nu_w = P_{S|W=w} = \alpha_w \nu_v|_{E_w}, \quad \alpha_w = \frac{\pi_v}{\pi_w}.$$

Every $s_0 \in E_w$ satisfies

$$\nu_v(B(s_0, R)) \geq 2^b R^k, \quad \nu_v(B(s_0, r_i)) \leq 2^{b+1} r_i^k \quad (i \geq 0).$$

Choose centers in E_w successively at distances greater than $2R$ from those already chosen, until their $2R$ -balls cover E_w . Their R -balls are disjoint and each has ν_v -mass at least $2^b R^k$; hence the

process stops with at most $(2^b R^k)^{-1}$ centers. Assign each point to one of its covering balls. Under the independent matrix G , conditioning also on this finite index adds at most $-b \log 2 - k \log R$ to the entropy. Within a ball, comparison with a Gaussian of covariance $(2R)^2 I_k$ adds at most $k \log R + Ck$, since $\mathbb{E}\|G(S - z)\|^2 \leq 4kR^2$ for its center z . Consequently

$$h_{\text{ind}}(GS \mid G, W = w) \leq -b \log 2 + Ck. \quad (48)$$

The entropies are finite by Lemma 2.3.

The same labels give, for all $z \in \mathbb{R}^d$ and $\rho > 0$,

$$\nu_w(B(z, \rho)) \leq C_0^d \min\{U_w \rho^k, \Gamma_w \rho^N\}, \quad U_w = \alpha_w 2^b, \quad \Gamma_w = \pi_w^{-1}. \quad (49)$$

For the first bound, choose a point of E_w in the ball when there is one, double the radius, and use a dyadic radius in $[2\rho, 4\rho]$ if $\rho \leq 1$. The upper ball bound at that point and restriction by α_w prove the assertion. For larger ρ , use total mass one and $U_w \geq 2^{-k}$. The second bound follows from $\nu_w \leq \Gamma_w \sigma$ and (30). Notice that $\alpha_w \geq 1$, $\Gamma_w \geq 1$, and $U_w \geq 2^{-k}$.

We next express the information difference in terms of exact projection densities. Let Q denote the independent law preserving (S, W) . Their (D, S) and (G, S) marginals agree after identifying the matrices. Hence $h_Q(GS \mid G) = h_P(DS \mid D)$. Proposition 2.4 gives

$$\mathcal{J}(W) - I_P(S; W \mid D, DS) = h_Q(GS \mid G, W) - h_P(DS \mid D, W) - I_P(S; A \mid W). \quad (50)$$

Here independence of B from (S, A, W) reduces $I_P(S; D \mid W)$ to $I_P(S; A \mid W)$. Lemma 2.3 makes the baseline and conditional entropies finite; it also gives the actual conditional image density. Thus (50) is an identity of finite quantities, with the displayed negative sign.

For a matrix D , let $p_{w,D}$ be a density of the image of ν_w under $s \mapsto Ds$. This denominator uses $\nu_w = P_{S|W=w}$ before conditioning on the actual rows. Contraction of

$$\mathbb{E}D(P_{S|A,W} \parallel \nu_W) = I_P(S; A \mid W)$$

under $s \mapsto Ds$, with B still independent, gives

$$-h_P(DS \mid D, W) - I_P(S; A \mid W) \leq \mathbb{E}_P \log p_{W,D}(DS). \quad (51)$$

The numerator after contraction is the true conditional image density given (D, W) . Its log ratio to $p_{W,D}$ is integrable: the relative entropy is finite, and the negative part of a log likelihood ratio has expectation at most $1/e$. Subtracting this ratio from the integrable actual log density also proves integrability of the right side.

Here we choose a density version that retains a bound at every fixed target s . For each positive integer a , let φ_δ^a be the centered Gaussian density of covariance $\delta^2 I_a$, and fix a sequence $\delta_t \downarrow 0$. For any probability measure $\nu \leq K\sigma$, define

$$F_\delta^\nu(g, s) = \int \varphi_\delta^k(g(u - s)) \nu(du), \quad p_\nu^b(g, y) = \liminf_{t \rightarrow \infty} \int \varphi_{\delta_t}^k(gu - y) \nu(du).$$

These functions are measurable. For each full-rank g , the Gaussian approximate identity converges to the image density at its Lebesgue points, so $p_\nu^b(g, \cdot)$ is a density version. It may be defined by a liminf on the exceptional set; this choice is useful at a fixed target. It is valid in (51), since the actual conditional image law is absolutely continuous.

There is a quantitative first-density passage under the independent law, before taking logarithms. For independent $G \sim \gamma_k$ and $S \sim \nu$, Gaussian integration and Tonelli give

$$\begin{aligned}\mathbb{E}_{G,S} F_\delta^\nu(G, S) &= (2\pi)^{-k/2} \iint (\|u - s\|^2 + \delta^2)^{-k/2} \nu(du) \nu(ds) \\ &\leq (2\pi)^{-k/2} \iint \|u - s\|^{-k} \nu(du) \nu(ds) < \infty.\end{aligned}\tag{52}$$

The last integral is finite uniformly in s : the diagonal has zero mass, and the centered cap bound on dyadic shells gives

$$\sup_{s \in S^{d-1}} \int \|u - s\|^{-k} \nu(du) \leq 1 + K 2^k \sum_{i \geq 0} 2^{-i(N-k)} < \infty.$$

At GS the approximate identity converges almost surely under the independent law, by absolute continuity of the image, Lebesgue differentiation, and Fubini. Fatou in (52) therefore gives

$$\mathbb{E}_{G,S} p_\nu^b(G, GS) \leq (2\pi)^{-k/2} \iint \|u - s\|^{-k} \nu(du) \nu(ds) < \infty.$$

This proves the positive logarithmic part directly. For the negative part compare with the standard Gaussian density ϕ on $\mathbb{R}^k$: $t(\log t)_- \leq 1/e$ bounds the negative part of $\log(p_\nu^b/\phi)$, and $\mathbb{E}\|GS\|^2 = k$ bounds the cross entropy with ϕ . This recovers independent log integrability for this density version. Actual log integrability is supplied by the contraction preceding (51). In what follows take $p_{w,D} = p_{\nu_w}^b(D, \cdot)$. The mollifiers here only specify and estimate densities at exact labels.

It remains to bound those densities under independent rows before changing the law of A . We claim that for every fixed w of positive probability and every $s \in S^{d-1}$,

$$\left[\mathbb{E}_A (\mathbb{E}_{B p_{\nu_w}^b}((A; B), (A; B)s))^q \right]^{1/q} \leq e^{Cd} U_w (1 + \log \Gamma_w),\tag{53}$$

where $A \sim \gamma_n$ and $B \sim \gamma_l$ are independent in this displayed expectation. We prove the uniform-in- δ version first, with

$$F_\delta(A, B, s) = \int \varphi_\delta^k((A; B)(u - s)) \nu_w(du).$$

For $h = u - s$,

$$\mathbb{E}_B \varphi_\delta^l(Bh) = (2\pi)^{-l/2} (\|h\|^2 + \delta^2)^{-l/2}.$$

Expanding the q -th power introduces $u_1, \dots, u_q$ and $h_i = u_i - s$. For their Gram matrix $K = (\langle h_i, h_j \rangle)_{i,j}$, Gaussian integration in the n shared rows gives exactly

$$\mathbb{E}_A \prod_{i=1}^q \varphi_\delta^n(Ah_i) = (2\pi)^{-nq/2} \det(K + \delta^2 I_q)^{-n/2}.$$

Using $\det(K + \delta^2 I_q) \geq \det K$ and Gram-Schmidt, we conclude

$$\mathbb{E}_A (\mathbb{E}_B F_\delta(A, B, s))^q \leq (2\pi)^{-kq/2} \int \prod_{i=1}^q \|u_i - s\|^{-l} \text{dist}(u_i - s, L_{i-1})^{-n} \nu_w^{\otimes q}(d\mathbf{u}),\tag{54}$$

where L_{i-1} is the span of $u_1 - s, \dots, u_{i-1} - s$, starting with $\{0\}$. An infinite upper bound is allowed at singular Gram matrices. All integrations so far are nonnegative.

We bound the corresponding one-point integral uniformly over s and every linear subspace L of dimension at most q . On a radial shell $R/2 < \|u - s\| \leq R$, $R = 2^{1-i}$, the part at distance at

most $2^{-a}R$ from L is covered by at most $(3 \cdot 2^a)^q$ balls of radius $2^{1-a}R$. This follows by covering the radius- R projection ball in L with a $2^{-a}R$ -net. By (49) its ν_w -mass is at most

$$C^d \min\{U_w 2^{-a(k-q)} R^k, \Gamma_w 2^{-a(N-q)} R^N\}.$$

In particular the set at distance zero has zero mass. On the band $2^{-a-1}R < \text{dist}(u-s, L) \leq 2^{-a}R$, the integrand $\|u-s\|^{-l} \text{dist}(u-s, L)^{-n}$ is at most $2^k 2^{an} R^{-k}$. Sum the transverse bands first. The positive gaps $k-q-n$ and $N-q-n$ make both sums geometric, so the integral on the radial shell is at most

$$C^d \min\{U_w, \Gamma_w R^{N-k}\}.$$

For the radial sum put $a_0 = N-k \geq d/3$. The sum of $\min\{1, (\Gamma_w/U_w) 2^{a_0(1-i)}\}$ has at most $2 + (\log_2(\Gamma_w/U_w))_+/a_0$ saturated terms followed by a geometric tail. Since $U_w \geq 2^{-k}$ and $\Gamma_w \geq 1$, this proves

$$\int \|u-s\|^{-l} \text{dist}(u-s, L)^{-n} \nu_w(du) \leq e^{Cd} U_w (1 + \log \Gamma_w). \quad (55)$$

Integrate (54) from its last point to its first, applying (55) each time. This proves the claimed bound uniformly in δ .

The liminf version now gives the exact fixed-target conclusion: for every fixed s ,

$$\mathbb{E}_B p_{\nu_w}^b((A; B), (A; B)s) \leq \liminf_{t \rightarrow \infty} \mathbb{E}_B F_{\delta_t}(A, B, s)$$

by Fatou in B . Raise to the q -th power and apply Fatou in A . The uniform moment bound proves (53), including at targets where a density version specified only almost everywhere would not have sufficed. This is the distinct fixed-target passage used in this comparison.

We finally pay for dependence of the actual rows. Their conditional relative entropies satisfy

$$\mathbb{E} D(P_{A|S,W} \| \gamma_n) = I_P(A; W | S) = I_P(A; V | S) \leq H(V). \quad (56)$$

Here A, S are marginally independent, and W adds only a function of (S, V) . Given (S, W, A) , B is still independent. Jensen in B , followed by (31) for the true $P_{A|S,W}$ relative to γ_n , uses (53) to give

$$\mathbb{E}_P \log p_{\nu_w}^b(D, DS) \leq \frac{H(V)}{q} + Cd + \mathbb{E}[\log \alpha_W + b \log 2 + \log(1 + \log \Gamma_W)]. \quad (57)$$

Finite average divergence ensures the required absolute continuity. Truncation permits the conditional inequality wherever a log expectation is negative infinite; the integrated cross entropy is finite by (51).

The $b \log 2$ term cancels (48) when substituted with (57) into (50)–(51). Moreover

$$\mathbb{E} \log \alpha_W = H(W | V), \quad \mathbb{E} \log(1 + \log \Gamma_W) \leq \log(1 + H(W)).$$

Removing the refinement costs another $H(W | V)$, since

$$\mathcal{J}(V) \leq \mathcal{J}(W), \quad I_P(S; W | D, DS) \leq I_P(S; V | D, DS) + H(W | V).$$

Thus the full quantitative budget is

$$\mathcal{J}(V) - I_P(S; V | D, DS) \leq Cd + \frac{H(V)}{q} + 2H(W | V) + \log(1 + H(W)). \quad (58)$$

Only now use $H(V) \leq d^2$, $q = \lfloor d/8 \rfloor$, and (47). They bound the right side by Cd , proving (46). $\square$

The applications of both comparisons, including their distinct residual-sphere constants, are proved in Subsection 8.4.

6 Actual and fresh projections through independent kernels

We compare the information in a block's incoming and ending states after revealing independent auxiliary measurements of the same signal. The comparison averages over the number of auxiliary rows.

We use the uniform-sphere experiment in Definition 1.1 and the entropy conventions of Section 2. Write $\gamma = N(0, I_d)$ for the law of a single row. All intermediate lemmas concern the fixed-seed experiment supplied by Lemma 1.2 and the Borel reduction in Lemma 8.1; in particular initialization remains jointly independent of the signal and all rows. Transitions may use fresh randomness, and the terminal output uses only its state and index. Auxiliary rows are analysis variables, independent of the joint signal and learner variables as specified below. Constants $c, C > 0$ are absolute and may increase. The estimates are for sufficiently large integer d .

For a nonnegative integer j , write $D'_j = (Z, ZS)$ for j fresh measurements. The matrix Z is independent of the joint vector consisting of the signal and all learner variables already generated; D'_0 is empty. Set

$$k_0 = \lceil d/4 \rceil, \quad k_1 = \lfloor d/2 \rfloor, \quad N_k = k_1 - k_0 + 1, \quad h_0 = \lfloor d/8 \rfloor, \quad q = \lfloor d/32 \rfloor. \quad (59)$$

For a finite state V , with fresh rows independent of (S, V) , define

$$J_k(V) = I(S; V \mid D'_k), \quad \bar{J}(V) = \frac{1}{N_k} \sum_{k=k_0}^{k_1} J_k(V). \quad (60)$$

These quantities are finite and nonnegative because they are at most $H(V)$.

Consider a block of $1 \leq m \leq h_0$ actual observations, with incoming state U and ending state V . Assume $H(V) \leq B < \infty$. The block rows $X_1, \dots, X_m$ are independent of (S, U) . The block transition produces V from U and the block data by a measurable kernel, conditionally independently of S .

Proposition 6.1 (Averaged information increase per block). *For the block above, with $1 \leq m \leq h_0$ and $H(V) \leq B$,*

$$\bar{J}(V) - \bar{J}(U) \leq Cm + \frac{m}{N_k} \left(C + \frac{B}{h_0} \right) + \frac{B}{q}. \quad (61)$$

When $B \leq d^2$, the right side is at most $C'd$ for an absolute constant C' .

Keeping V as produced by the actual block, replace fresh auxiliary rows one at a time by rows the learner actually saw. Conditioning on the ending state can bias an actual row; the total row-information cost is at most $H(V)$. Independent random kernels control the accompanying entropy cost, and averaging over the auxiliary count reduces the difference of successive conditional entropies to two endpoint terms.

6.1 Residual geometry and finite label entropies

The following residual-sphere estimates justify the entropy subtractions and will also normalize the row comparison.

Lemma 6.2 (Residual-sphere estimates). *Let $D_j = (X, XS)$ consist of j independent Gaussian measurements, where $0 \leq j < d$, and put $n = d - j$. Conditional on D_j , the signal is uniform on a sphere of radius R in the affine solution space. Marginally,*

$$R^2 \stackrel{\text{law}}{=} \frac{G_n}{G_n + G_j}, \quad \mathbb{E}R^{-2} = 1 + \frac{j}{n-2} \quad (n > 2), \quad (62)$$

where G_n, G_j are independent chi-squared variables and $G_0 = 0$.

If $j \leq 3d/4$, the entropy of one further label given its row and D_j lies between two absolute constants, and its conditional log density is absolutely integrable. Joint label entropies, and the conditional label entropies obtained by also conditioning on a finite state V , are finite whenever the total number of rows is at most $3d/4$. The associated conditional log densities are absolutely integrable. Removing V can increase any such joint entropy by at most $H(V)$.

Let σ_R be uniform probability on a sphere of radius $R > 0$ in an n -dimensional affine space. A Euclidean ball of radius a with arbitrary center has σ_R -mass at most $(2a/R)^{n-1}$ when $2a/R \leq 1/2$. For $n \geq 3$,

$$\sup_{s \in \text{supp } \sigma_R} \int \frac{\sigma_R(dt)}{\|t - s\|} \leq \frac{C}{R}. \quad (63)$$

Proof. The rows have full rank almost surely. Their labels determine the row-space component of S . Rotational invariance makes the remaining direction uniform in the kernel, conditionally on its length and the data. Representing a uniform sphere point as a standard Gaussian vector divided by its norm proves the ratio law in (62). For $n > 2$, independence and the chi-squared density give $\mathbb{E}G_n^{-1} = 1/(n-2)$ and $\mathbb{E}G_j = j$, proving the inverse moment.

Let P be projection onto the kernel and let x be a new row. Conditional on D_j, x , the new label is a fixed shift plus

$$R\|Px\|U_1,$$

where U is uniform on S^{n-1} . The density of U_1 is proportional to $(1 - u^2)^{(n-3)/2}$ on $(-1, 1)$. Its normalizing integral is at least $c/\sqrt{n}$, by integrating over $|u| \leq 1/\sqrt{n}$. For the dimensions in use this density is at most $C\sqrt{n}$, and hence

$$h(U_1) \geq -\log(C\sqrt{n}).$$

Its log density is integrable: near an endpoint this is the integrability of $t^\alpha |\log t|$ for $\alpha > -1$.

Also $\|Px\|^2$ is chi-squared with n degrees of freedom. For $j \leq 3d/4$, the inverse moments of R and $\|Px\|/\sqrt{n}$ are bounded by absolute constants. Jensen's inequality gives, for example,

$$\mathbb{E} \log R \geq -\frac{1}{2} \log \mathbb{E} R^{-2} \geq -C, \quad \mathbb{E} \log (\|Px\|/\sqrt{n}) \geq -C.$$

Ordinary positive moments control the positive logarithmic parts. The entropy formula under scaling now gives an absolute lower bound for the conditional entropy of the next label. The upper bound is $\frac{1}{2} \log(2\pi e)$, the entropy of its marginal standard normal law. These calculations also give absolute integrability of the conditional log density. Applying the chain rule in any order gives the same conclusion for all ordered subsets of labels with the stated row budget. Extra independent rows may be conditioned on before V is added, since they do not change these baseline distributions.

Adding the finite state reduces a joint label entropy by a conditional mutual information at most $H(V)$. To check absolute integrability as well, let $r = dQ/dP$ be the density ratio of the conditional joint law to its product-of-conditionals reference law. The negative part of $\int \log r dQ$ is at most $1/e$, because $-r \log r \leq 1/e$ for $0 < r \leq 1$. Finite relative entropy therefore also makes the positive part integrable. The log density after conditioning on V is the baseline log density plus this integrable log ratio. All label-entropy chain rules just used are consequently between finite quantities.

For $n = 1$ the cap bound is the trivial bound one. For $n \geq 2$, first scale to $R = 1$ and center the ball at a sphere point. If its radius is $a \leq 1/2$, its projection onto the tangent coordinate plane lies in an $(n-1)$ -ball of radius a . The graph area factor there is at most 2, while the whole sphere has area at least twice the volume of the unit $(n-1)$ -ball. The cap has mass at most a^{n-1} . For an

arbitrary center, choose a sphere point in the intersection, if one exists, and double the radius. This proves the displayed arbitrary-center bound.

Finally, for a sphere point s , integrate the centered cap estimate against $t^{-2} dt$ for $0 < t \leq R/2$; the integral is bounded by

$$R^{-(n-1)} \int_0^{R/2} t^{n-3} dt \leq C/R$$

when $n \geq 3$. The part with $t > R/2$ is at most C/R using total mass one. The layer-cake formula for $\|t - s\|^{-1}$ proves (63). $\square$

6.2 A block reduction and its row-information cost

For this block, write

$$X_{\leq i} = (X_1, \dots, X_i), \quad Y_{\leq i} = X_{\leq i} S, \quad D_i = (X_{\leq i}, Y_{\leq i}), \quad L_i = (V, D_i)$$

for $0 \leq i \leq m$, with the evident empty tuples at $i = 0$. The rows are Gaussian before conditioning on V ; their conditional laws after V is known need not be Gaussian.

For an independent sequence of fresh rows $z_1, z_2, \dots$, define the successive conditional entropies

$$a_j^{(i)} = h(z_j S \mid z_j, D'_{j-1}, L_i), \quad j \geq 1, \quad i + j \leq 3d/4. \quad (64)$$

All the entropies used below have a total row count at most $3d/4$, so Lemma 6.2 applies.

Lemma 6.3 (One-row hybrid identity). *For $k_0 \leq k \leq k_1$ and $0 \leq i \leq m$, let $Z^{(i,k)}$ be a fresh matrix of $k - i$ rows and define*

$$G_i(k) = h((Y_{\leq i}, Z^{(i,k)} S) \mid X_{\leq i}, Z^{(i,k)}, V) + I(S; X_{\leq i} \mid V).$$

For $i < m$, put $x = X_{i+1}$, $Y = xS$, $L = L_i$, and $r = k - i - 1$. With $F_r = (Z, ZS)$ denoting common fresh data of size r ,

$$G_i(k) - G_{i+1}(k) = a_{r+1}^{(i)} - [h(Y \mid x, F_r, L) + I(S; x \mid F_r, L)]. \quad (65)$$

The actual-row divergence costs are

$$K_i := \mathbb{E} D_{\text{KL}}(\mathcal{L}(X_{i+1} \mid S, L_i) \parallel \gamma) = I(X_{i+1}; V \mid S, X_{\leq i}), \quad (66)$$

$$\sum_{i=0}^{m-1} K_i = I(X_1, \dots, X_m; V \mid S) \leq H(V) \leq B. \quad (67)$$

Finally,

$$J_k(V) \leq J_k(U) + G_0(k) - G_m(k). \quad (68)$$

Proof. We first relate $G_i(k)$ to conditional information without subtracting differential entropies of the singular signal. Suppose $D = (A, AS)$ has the marginal law of k independent Gaussian measurements, although its rows may have helped produce V . The conditional relative-entropy chain rule gives

$$I(S; V \mid D) - I(S; V) = I(D; V \mid S) - I(D; V).$$

Every term here is finite, being bounded by $H(V)$. The labels are determined by A, S , and A is marginally independent of S . Expanding the finite terms involving V therefore yields

$$I(S; V \mid D) = I(S; V) - h(AS \mid A) + h(AS \mid A, V) + I(S; A \mid V). \quad (69)$$

The label entropies are finite by Lemma 6.2. In a hybrid, the fresh rows contribute zero to the last mutual information. All hybrids have the same baseline entropy $h(AS | A)$. Thus

$$J_k(V) - I(S; V | D_m, D'_{k-m}) = G_0(k) - G_m(k).$$

Given D_m, D'_{k-m} , the block transition is a Markov kernel from U to V , conditionally independently of S . Conditional data processing gives

$$I(S; V | D_m, D'_{k-m}) \leq I(S; U | D_m, D'_{k-m}) = J_k(U).$$

The equality holds because these k rows are fresh relative to (S, U) . This proves (68).

For the one-row identity, realize the two adjacent hybrids using the same r fresh rows Z , one further fresh row for $G_i(k)$, and the actual row x for $G_{i+1}(k)$. Expand the labels in the order $Y_{\leq i}, ZS$, last label. Before the last row is added the common part is

$$h(Y_{\leq i} | X_{\leq i}, V) + I(S; X_{\leq i} | V) + h(ZS | Z, L).$$

The remaining term for $G_i(k)$ is $a_{r+1}^{(i)}$. For $G_{i+1}(k)$, conditioning the first two label groups on the actual row subtracts, respectively,

$$I(Y_{\leq i}; x | X_{\leq i}, V) \quad \text{and} \quad I(ZS; x | Z, L).$$

Its new row-information term is initially $I(S; x | X_{\leq i}, V)$. The first subtraction leaves $I(S; x | L)$, because the old labels are determined by $S, X_{\leq i}$. The second leaves $I(S; x | F_r, L)$: the fresh matrix Z is independent, and ZS is determined by S, Z . The last label then contributes $h(Y | x, F_r, L)$. This proves (65) and retains both earlier-label subtractions.

Conditional on $S, X_{\leq i}$, the unconditioned law of X_{i+1} is γ , and the labels in L_i are redundant. The definition of conditional mutual information proves (66). Summing it is the conditional chain rule for $X_1, \dots, X_m$. The resulting information is at most $H(V)$, proving (67). In particular the conditional actual-row laws have finite divergence from γ almost surely. $\square$

The identity has isolated the issue caused by conditioning on V . We next lower-bound the bracket in (65). Its reference posterior will exclude the actual row, and the change from a Gaussian row to the actual row will be charged through K_i .

6.3 Independent kernels and the biased-row comparison

Two signals giving the same fresh labels differ by a vector in the fresh matrix's kernel. The posterior will select such directions nonuniformly, so we need an inverse-volume estimate that holds for every selection rule allowed below.

Lemma 6.4 (Inverse volumes for independent kernels). *Let r, q be integers satisfying*

$$d/8 \leq r < d, \quad 1 \leq q \leq r/2, \quad r - q \geq 2.$$

Generate independent standard Gaussian r -by- d matrices $Z_1, \dots, Z_q$. After generating Z_j , select a unit vector $u_j \in \ker Z_j$ by a probability kernel that may depend on Z_j and all preceding matrices and selections, but not on future matrices. Then

$$\mathbb{E} \text{vol}(u_1, \dots, u_q)^{-1} \leq C^q, \tag{70}$$

uniformly over all such selection kernels. In particular the volume is positive almost surely.

Proof. Fix a previous span P of dimension $p < q$, and let E be the row space of the next matrix. When $p > 0$, set

$$\xi(P, E) = \inf_{\substack{v \in P \\ \|v\|=1}} \|P_E v\|,$$

and set $\xi = 1$ for $p = 0$. An operator and its adjoint have the same norm, so $\|P_P|_{E^\perp}\| = \|P_{E^\perp}|_P\|$. For $p > 0$, taking complements of their squared norms shows, for every unit $u \in E^\perp$, that

$$\text{dist}(u, P)^2 \geq 1 - \|P_P|_{E^\perp}\|^2 = \inf_{\substack{v \in P \\ \|v\|=1}} \|P_E v\|^2 = \xi(P, E)^2. \quad (71)$$

For $p = 0$, both the distance and ξ equal one, so all the bounds below are immediate. We may therefore assume $p > 0$ in the small-ball calculation.

The new E is a uniform r -dimensional subspace independent of the preceding selections. For a fixed unit v and $0 < t \leq 1/4$,

$$\mathbb{P}\{\|P_E v\| \leq 2t\} \leq (Ct)^r. \quad (72)$$

Indeed $\|P_E v\|$ has the distribution of $\|g\|/(\|g\|^2 + \|g'\|^2)^{1/2}$, with independent standard Gaussian vectors of dimensions r and $d - r$. The event forces $\|g\| \leq Ct\|g'\|$. The Gaussian integral, restricted to the ball of radius $\sqrt{r}$, gives $(2\pi)^{r/2} \geq e^{-r/2} v_r r^{r/2}$, hence $v_r \leq (C/\sqrt{r})^r$. Conditional on g' , this volume estimate and the Gaussian density bound make its probability at most $(Ct\|g'\|/\sqrt{r})^r$. For $z \geq 0$,

$$z^{r/2} \leq (2r/e)^{r/2} e^{z/4}, \quad \mathbb{E} e^{\|g'\|^2/4} = 2^{(d-r)/2}.$$

Since $r \geq d/8$, these estimates give $\mathbb{E}\|g'\|^r \leq (C\sqrt{d})^r$. Averaging proves (72).

A maximal t -separated subset of the unit sphere in P is a t -net with at most $(1 + 2/t)^p \leq (3/t)^p$ points, by packing disjoint balls of radius $t/2$ in the ball of radius $1 + t/2$. Projection is a contraction, so $\xi \leq t$ puts some net point at projection distance at most $2t$. By (72),

$$\mathbb{P}\{\xi \leq t\} \leq (3/t)^p (Ct)^r \leq (C_0 t)^{r-p}, \quad 0 < t \leq 1/4. \quad (73)$$

Here C_0 is absolute because $p < q \leq r/2$. Choose a fixed $t_0 \leq 1/4$ such that $C_0 t_0 \leq 1/2$. Since $r - p > 1$, integration of (73) gives

$$\mathbb{E}\xi^{-1} = 1 + \int_0^1 \mathbb{P}\{\xi < t\} \frac{dt}{t^2} \leq C.$$

For $t \geq t_0$, the trivial probability bound suffices. The estimate is uniform in the preceding span P .

Gram-Schmidt expresses the volume as the product of distances from successive preceding spans. Conditional on all previous selections, the $p = 0$ observation and (71) bound the reciprocal of the next distance by ξ^{-1} , whatever vector the new kernel selects. The new row space is still independent of that history. Iterated conditional expectation therefore bounds the inverse-volume product by C^q , and also shows that none of the distances vanishes with positive probability. $\square$

Return to a fixed i, k in Lemma 6.3. Thus $x = X_{i+1}$, $L = (V, D_i)$, $r = k - i - 1$, and $F_r = (Z, ZS)$ is fresh. Define the reference posterior and its reciprocal-distance scale by

$$\mu = \mathcal{L}(S \mid L, F_r), \quad A(s, L, F_r) = \int \frac{\mu(ds')}{\|s' - s\|}. \quad (74)$$

The posterior μ includes the ending state but does not condition on the actual row x . This distinction is essential: we will compare the actual conditional law of x with a Gaussian row.

Lemma 6.5 (A row biased by the ending state). *In the preceding experiment, $A(S, L, F_r) \geq 1/2$ almost surely and $\mathbb{E}A(S, L, F_r) < \infty$ for each fixed d . For almost every Gaussian row x , the projection of μ by $s \mapsto xs$ has a Lebesgue density $p_{\mu,x}$. With $q = \lfloor d/32 \rfloor$,*

$$h(xS \mid x, L, F_r) + I(S; x \mid L, F_r) \geq -\mathbb{E} \log p_{\mu,x}(xS), \quad (75)$$

$$\mathbb{E} \log \frac{p_{\mu,x}(xS)}{A(S, L, F_r)} \leq C + \frac{K_i}{q}. \quad (76)$$

Both expectations use the actual row law. The logarithms in these displays are absolutely integrable.

Proof. Conditional on D_i, F_r , before V is imposed, the signal is uniform on a residual sphere of affine dimension

$$n = d - i - r = d - k + 1 > 2.$$

For a state v with $\pi_v = \mathbb{P}\{V = v \mid D_i, F_r\} > 0$, its posterior density relative to the uniform residual law is at most $1/\pi_v$. This follows directly from Bayes' formula, since the conditional probability of $V = v$ given the signal and these data is at most one. Lemma 6.2 therefore gives

$$A(s, L, F_r) \leq \frac{C}{R\pi_v}$$

for every point s on the residual sphere. Average first over s and v . The factor π_v cancels, and the sum has at most W terms if V has at most W values. Consequently

$$\mathbb{E}A(S, L, F_r) \leq CW \mathbb{E}R^{-1} < \infty. \quad (77)$$

The inverse moment in (62) proves the last inequality. Every pairwise distance on the unit sphere is at most two, giving $A \geq 1/2$. In particular $\log A$ is absolutely integrable.

The posterior is nonatomic and is absolutely continuous relative to the uniform residual law. For every row whose projection into the residual kernel is nonzero, the projection of that uniform law has a Lebesgue density, by the one-coordinate formula in Lemma 6.2. The exceptional rows have Gaussian measure zero. Absolute continuity passes to the projected posterior, proving the assertion about $p_{\mu,x}$.

The information cost of conditioning this posterior on x is finite. Indeed x and F_r are conditionally independent given S, L , so the conditional chain rule yields

$$I(S; x \mid L) - I(S; x \mid L, F_r) = I(x; F_r \mid L) \geq 0.$$

Also the chain rule for the relative entropy in K_i gives

$$K_i = I(S; x \mid L) + \mathbb{E}D_{\text{KL}}(\mathcal{L}(x \mid L) \parallel \gamma).$$

Thus

$$I(S; x \mid L, F_r) \leq I(S; x \mid L) \leq K_i. \quad (78)$$

The conditional actual law of S given x, L, F_r is therefore dominated by μ almost surely. Projecting by $s \mapsto xs$ contracts its relative entropy, so the actual projected law is dominated by the reference projected law with expected divergence at most $I(S; x \mid L, F_r)$. Lemma 6.2 supplies an integrable log density for the actual conditional label. Applying the log-density-ratio argument from that lemma to the finite projected divergence shows that $\log p_{\mu,x}(xS)$ is absolutely integrable as well. We may now expand the projected divergence:

$$-\mathbb{E} \log p_{\mu,x}(xS) - h(xS \mid x, L, F_r) \leq I(S; x \mid L, F_r).$$

This is (75).

It remains to prove (76). Fix a typical pair (s, L) under the joint law of (S, L) . Generate q independent copies $Z_1, \dots, Z_q$ of the r fresh rows, and put $F^{(j)} = (Z_j, Z_j s)$. Let μ_j, A_j be the corresponding posterior and scale from (74). Initially let $x \sim \gamma$ be independent of all these matrices. We claim

$$\mathbb{E}_{Z_1, \dots, Z_q, x} \prod_{j=1}^q \frac{p_{\mu_j, x}(xs)}{A_j} \leq C^q. \quad (79)$$

For $\delta > 0$, replace the density in the j -th factor by its symmetric slab average

$$p_{\mu_j, x}^\delta(xs) = \frac{1}{2\delta} \mu_j\{s' : |x(s' - s)| \leq \delta\}.$$

Draw s'_j separately from μ_j , and write $\Delta_j = s'_j - s$, $u_j = \Delta_j / \|\Delta_j\|$. The posteriors are nonatomic, so the differences are nonzero almost surely. Conditional on these differences, the vector $(x\Delta_1, \dots, x\Delta_q)$ is Gaussian. Its density is bounded by

$$(2\pi)^{-q/2} \prod_{j=1}^q \|\Delta_j\|^{-1} \text{vol}(u_1, \dots, u_q)^{-1}; \quad (80)$$

we may use $+\infty$ for a singular tuple. This is the usual Gaussian density bound with covariance matrix $(\langle \Delta_j, \Delta_\ell \rangle)_{j, \ell}$. Its probability in $[-\delta, \delta]^q$ is at most (80) times $(2\delta)^q$.

After division by $\prod_j A_j$, nonnegative integration converts the resulting bound into $(2\pi)^{-q/2}$ times an inverse-volume expectation. Conditional on each Z_j , the selected point in that expectation has the probability law

$$\nu_j(ds') = \frac{\mu_j(ds')}{\|s' - s\| A_j}. \quad (81)$$

Each law is normalized by its own A_j . Its total mass is one for every typical Z_j , so the marginal law of the matrices remains the product of their original Gaussian laws. In particular the matrices have not been tilted. Both s and s'_j satisfy the labels $Z_j s$; therefore $u_j \in \ker Z_j$. These selections, made separately conditional on their matrices and (s, L) , satisfy the hypotheses of Lemma 6.4. Indeed

$$r = k - i - 1 \geq k_0 - h_0 \geq d/8, \quad q \leq r/2, \quad r - q \geq 2$$

for sufficiently large d . The lemma bounds the smoothed product expectation by C^q , uniformly in δ .

Here is a density version that makes passage to exact evaluations legitimate. In the unbiased one-copy experiment, x is independent of (S, L, F_r) , so conditional on L, F_r, x , the signal has law μ . The one-dimensional Lebesgue differentiation theorem applied to the integrable density $p_{\mu, x}$ shows that xS is almost surely a Lebesgue point and that the symmetric slab averages converge to $p_{\mu, x}(xs)$. Each slab average is jointly measurable in its kernel parameters and evaluation point. Taking its limit along $\delta = 2^{-n}$, and assigning zero when no finite limit exists, therefore gives a jointly measurable density version that agrees at these evaluations. Fubini's theorem gives the same assertion for almost every fixed (s, L) . For such a pair it holds simultaneously for all q copies, since q is finite. Fatou's lemma now proves (79).

Change only x from its unbiased law to its actual conditional law given $S = s, L$. The Z_j remain independent of it and of one another. The divergence from the unbiased product experiment is exactly $D_{\text{KL}}(\mathcal{L}(x | s, L) \| \gamma)$. For probability laws $Q \ll P$ and a nonnegative function g , the entropy variational inequality gives

$$\mathbb{E}_Q \log g \leq D_{\text{KL}}(Q \| P) + \log \mathbb{E}_P g. \quad (82)$$

For positive bounded g this follows from Jensen's inequality applied to $g/(dQ/dP)$; truncation gives the extended form. This is the standard variational identity recorded in [3, Equations (1.1)–(1.2)]. Apply it to the product in (79). Finite row divergence transfers the unbiased almost-sure density statements to the actual law. The logarithms are integrable by (75) and (77). The expectation of the log product is q times the one-copy expectation, since the q fresh copies have the same marginal law. Divide by q and average over (S, L) . Equation (66) then gives (76). $\square$

The independent-kernel estimate bounds the actual projected density at the scale A . To compare this scale to a fresh conditional entropy, we use two fresh rows after the r common rows. The second of the two is the next slope beyond the one already present in the hybrid identity.

6.4 A reciprocal-distance tilt and two fresh rows

Lemma 6.6 (The next fresh entropy slope). *For the posterior and scale in (74),*

$$a_{r+2}^{(i)} \leq -\mathbb{E} \log A(S, L, F_r) + C. \quad (83)$$

Proof. Write $C_* = (L, F_r)$. Conditional on C_* , the signal has law μ . For an integer $N \geq 1$, define

$$w_N(s, t) = \min\{N, \|t - s\|^{-1}\}, \quad A_N(s, C_*) = \int w_N(s, t) \mu(dt),$$

where $w_N(s, s) = N$. Given $S = s, C_*$, draw an auxiliary point S^* from the probability kernel

$$\nu_s^N(dt) = \frac{w_N(s, t)}{A_N(s, C_*)} \mu(dt).$$

Distances on the unit sphere are at most two, so $1/2 \leq A_N \leq N$ and $d\nu_s^N/d\mu \leq 2N$. The marginal law of S^* need not equal μ .

Compare the joint conditional law of (S, S^*) with $\mu \otimes \mu$. Its divergence is

$$\mathbb{E} \log \frac{w_N(S, S^*)}{A_N(S, C_*)}.$$

The relative-entropy chain rule splits this into $I(S; S^* \mid C_*)$ plus the divergence of the actual S^* marginal from the comparator μ . Hence

$$\begin{aligned} I(S; S^* \mid C_*) &\leq \mathbb{E} \log \frac{w_N(S, S^*)}{A_N(S, C_*)} \\ &\leq -\mathbb{E} \log \|S - S^*\| - \mathbb{E} \log A_N(S, C_*). \end{aligned} \quad (84)$$

The comparator divergence is finite, at most $\log(2N)$. The points differ almost surely because μ is nonatomic. Their log distance is absolutely integrable: its positive part is at most $\log 2$; under $\mu \otimes \mu$ its negative part is bounded by an integrable multiple of $\|S - S^*\|^{-1}$ by (77), and the tilted density is at most $2N$. Thus both inequalities in (84) are between well-defined finite quantities.

After drawing S^* , generate independent fresh Gaussian rows x_1, x_2 and labels $b_1 = x_1 S, b_2 = x_2 S$. Before S^* is added to the conditioning, Lemma 6.2 gives a joint conditional density with an integrable logarithm for these labels, since $i + r + 2 = k + 1 \leq 3d/4$. Fresh-row independence and data processing give

$$\begin{aligned} I(b_2; S^* \mid b_1, x_1, x_2, C_*) &\leq I((b_1, b_2); S^* \mid x_1, x_2, C_*) \\ &\leq I(S; S^* \mid C_*). \end{aligned} \quad (85)$$

This is finite. The log-density-ratio argument from Lemma 6.2 consequently gives integrable conditional log densities after S^* is added, both for the joint labels and for the first label. Their entropy chain rule justifies the conditional entropy of the second label used next.

Given b_1, x_1, x_2, C_*, S^* , compare the density of b_2 with the Cauchy density centered at $x_2 S^*$ and with scale $|b_1 - x_1 S^*|$. The scale is positive almost surely. With $\Delta = S - S^*$, the expected negative log of this reference density at b_2 is

$$\begin{aligned} \log \pi + \mathbb{E} \log |x_1 \Delta| + \mathbb{E} \log \left(1 + \frac{(x_2 \Delta)^2}{(x_1 \Delta)^2} \right) \\ = \mathbb{E} \log \|\Delta\| + C_1, \end{aligned} \quad (86)$$

where C_1 is absolute. Conditional on nonzero Δ , the two projections divided by $\|\Delta\|$ are independent standard normals G_1, G_2 . The first logarithm is integrable because $\mathbb{E} |\log |G_1|| < \infty$. For the ratio term,

$$\log(1 + (G_2/G_1)^2) = \log(G_1^2 + G_2^2) - \log G_1^2,$$

and both logarithms on the right are integrable by the one- and two-dimensional chi-squared densities. The cross-entropy inequality therefore gives

$$h(b_2 \mid b_1, x_1, x_2, C_*, S^*) \leq \mathbb{E} \log \|\Delta\| + C_1.$$

Removing S^* from the conditioning in this entropy bound and using (84)–(85) cancels $\mathbb{E} \log \|\Delta\|$. We obtain

$$h(b_2 \mid b_1, x_1, x_2, C_*) \leq -\mathbb{E} \log A_N(S, C_*) + C_1.$$

The left side is exactly $a_{r+2}^{(i)}$ and is independent of N . Finally $A_N \uparrow A$ and $\log A_N \geq -\log 2$. Monotone convergence, with the integrability of $\log A$ already proved, yields (83). $\square$

6.5 Averaging over the number of side measurements

The row comparison and the two-row coupling now leave only a difference of consecutive fresh entropy slopes. Averaging over consecutive values of k makes this difference telescope.

Proof of Proposition 6.1. Equations (75) and (76) lower-bound the bracket in (65) by $-\mathbb{E} \log A - C - K_i/q$. Equation (83) then gives

$$h(xS \mid x, L, F_r) + I(S; x \mid L, F_r) \geq a_{r+2}^{(i)} - C - K_i/q.$$

Since $r = k - i - 1$, the hybrid identity becomes

$$G_i(k) - G_{i+1}(k) \leq a_{k-i}^{(i)} - a_{k-i+1}^{(i)} + C + K_i/q. \quad (87)$$

For fixed i , the fresh observations are exchangeable conditional on L_i . Relabel two consecutive fresh observations and then condition on the additional earlier one. Conditional entropy can only decrease, so the successive slopes $a_j^{(i)}$ are nonincreasing in their finite range. Also

$$a_j^{(i)} \leq \frac{1}{2} \log(2\pi e)$$

by comparison with the marginal standard normal label.

For $i + j \leq k_1 + 1$, a useful lower bound is

$$a_j^{(i)} \geq -C - B/h_0. \quad (88)$$

To prove it, use monotonicity to bound $a_j^{(i)}$ below by the average of the next h_0 slopes. Their sum is the joint entropy of the next h_0 fresh labels given their rows, D'_j , and L_i . Removing V from that conditioning increases the entropy by at most $H(V) \leq B$. Without V , the actual prefix and all fresh rows have the ordinary independent-row law. Each successive baseline label entropy is at least $-C$ by Lemma 6.2, since the total number of rows is at most

$$i + j + h_0 \leq k_1 + 1 + h_0 \leq 3d/4$$

for sufficiently large d . The sum is therefore at least $-Ch_0 - B$, proving (88).

Average (87) over $k_0 \leq k \leq k_1$. For each i , its slope term is exactly

$$\frac{1}{N_k} \sum_{k=k_0}^{k_1} (a_{k-i}^{(i)} - a_{k-i+1}^{(i)}) = \frac{a_{k_0-i}^{(i)} - a_{k_1-i+1}^{(i)}}{N_k} \leq \frac{C + B/h_0}{N_k}.$$

Now sum over $i = 0, \dots, m-1$, use $\sum_i K_i \leq B$ from (67), and apply (68). This gives (61). Finally $N_k, h_0, q \geq cd$ and $m \leq d/8$, so $B \leq d^2$ makes the displayed block cost at most $C'd$. $\square$

Subsection 8.5 combines this block estimate with the stopping clock and the terminal residual-sphere bound.

7 A finite Gaussian-fiber comparison

We now bound the information retained after a block of exact Gaussian measurements by comparing two ways to average the state likelihood along a measurement fiber. The comparison measure is finite but need not have mass one. Its marginal is an inverse-distance kernel involving the rows actually used by the transition. A common local-mass quantity then connects that marginal to a projection independent of the new state.

We retain the measure and entropy conventions of Section 2. Constants denoted by K are positive and absolute and may increase. The local row counts are

$$n = \lfloor d/10 \rfloor, \quad k = \lfloor d/3 \rfloor, \quad \ell = \lfloor d/10 \rfloor. \quad (89)$$

Here n is the number of rows read in one block, k is the number of rows in an auxiliary projection, and ℓ is an integer moment order. We take d sufficiently large that $\ell \geq 2$ and $k - \ell - n \geq d/10$.

7.1 Projection fibers and conditional information

We first record the geometric facts used both in the block estimate and at the terminal state. If σ_m is uniform probability on S^{m-1} , then for $m \geq 2$, $z \in S^{m-1}$, and $0 < r \leq 1/2$,

$$\sigma_m(B(z, r)) \leq r^{m-1}. \quad (90)$$

This is the centered-ball estimate of Lemma 2.1 in dimension m .

For a full-row-rank $b \times d$ matrix C , where $1 \leq b \leq d-2$, put

$$P_C = C^\top (CC^\top)^{-1} C, \quad x_C(z) = C^\top (CC^\top)^{-1} z.$$

Write $S(H) = \{u \in H : \|u\| = 1\}$ for the unit sphere of a subspace H . The conditional law of $S \sim \sigma$ given $CS = z$, denoted by $\sigma_{C,z}$, fixes $P_C S = x_C(z)$ and is uniform on the remaining sphere

$$x_C(z) + \sqrt{1 - \|x_C(z)\|^2} S(\ker C), \quad \|x_C(z)\| < 1.$$

This formula specifies a Borel probability kernel on the indicated set. For example, the uniform direction in $\ker C$ can be realized as $(I - P_C)Z/\|(I - P_C)Z\|$ with Z a standard Gaussian vector; the map is Borel away from a null set. Set the kernel to a fixed point mass off the full-rank interior set. To verify the formula, write a uniform sphere point as a standard Gaussian vector divided by its length. The squared lengths in the row space and its orthogonal complement are independent chi-squared variables with b and $d - b$ degrees of freedom before division. The first divided by their sum has the beta distribution with parameters $b/2$ and $(d - b)/2$, and the two directions are independent and uniform. Consequently the row-space projection, in orthonormal coordinates, has density

$$c_{d,b}(1 - \|x\|^2)^{(d-b-2)/2} \quad (\|x\| < 1), \quad c_{d,b} = \frac{\Gamma(d/2)}{\pi^{b/2}\Gamma((d-b)/2)}. \quad (91)$$

The density is zero outside the unit ball. In particular, the conditional statement holds for almost every observation. Since $\|x_C(z)\|^2 = z^\top(CC^\top)^{-1}z$, the Lebesgue density p_C of CS is

$$p_C(z) = \begin{cases} \frac{c_{d,b}}{\det(CC^\top)^{1/2}}(1 - \|x_C(z)\|^2)^{(d-b-2)/2}, & \|x_C(z)\| < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (92)$$

Boundary values do not affect any of the integrals below.

For $b = k$, the exponent in (91) is nonnegative. Integrating its normalization over the ball of radius $1/2$ shows that its supremum times the volume of the unit k -ball is at most $2^k(4/3)^{d/2}$. Thus, for every ball D of radius r in the row space,

$$\Pr\{P_CS \in D\} \leq 2^k(4/3)^{d/2}r^k \leq e^{Kd}r^k. \quad (93)$$

The estimate is for the unweighted uniform measure σ .

Let V take values in a finite set $\mathcal{V}$ and be jointly distributed with S . Choose Borel versions of the state likelihoods

$$f_v(s) = \Pr\{V = v \mid S = s\}, \quad \bar{f}_v(C, z) = \int f_v(t) d\sigma_{C,z}(t).$$

Modify the first family on a Borel σ -null set, if needed, so that $f_v \geq 0$ and $\sum_v f_v = 1$ everywhere. These are likelihoods, not probability densities normalized separately for each state. When C has k independent standard Gaussian rows and is independent of (S, V) , disintegration identifies $\bar{f}_v(C, z)$ with $\Pr\{V = v \mid C, CS = z\}$. The conditional relative-entropy chain rule therefore gives

$$\begin{aligned} \Phi(V) &:= I(S; V \mid C, CS) = \mathbb{E} \log \frac{f_V(S)}{\bar{f}_V(C, CS)} \\ &= H(V \mid C, CS) - H(V \mid S). \end{aligned} \quad (94)$$

Here H denotes Shannon entropy in nats. For the standard-Borel form of this chain rule, see [1, Section 3.3]. Nonnegativity of conditional relative entropy and the displayed difference give $0 \leq \Phi(V) \leq \log |\mathcal{V}|$. Terms on zero-probability state or fiber events are omitted.

Proposition 7.1 (Information through a Gaussian block). *Let $S \sim \sigma$, let V be finite-valued, and let A have n independent standard Gaussian rows, independently of (S, V) . Suppose W takes at most N values, where $N \geq 1$ is an integer, and is obtained from (V, A, AS) by a Borel transition kernel. If $\log N \leq d^2$, then for the dimensions in (89) and all sufficiently large d ,*

$$\Phi(W) \leq \Phi(V) + Kd. \quad (95)$$

The auxiliary matrix in each occurrence of Φ is independent of the pair consisting of the signal and the state in that occurrence.

The proof first compares the real law to a finite measure. We next identify the latter's signal/state marginal by a two-point Borel measure identity. Finally, two estimates with different geometric proofs compare that marginal and the fresh-projection likelihood to one local mass.

7.2 The finite measure and its exact marginal

Write the transition probabilities as $h_{vw}(A, AS)$. Define

$$g_w(A, t) = \sum_v f_v(t) h_{vw}(A, At), \quad f_w(t) = \mathbb{E}_A g_w(A, t). \quad (96)$$

Thus $0 \leq g_w \leq 1$, $0 \leq f_w \leq 1$, and the sums over w of each of these functions are one. The function f_w is precisely the new state's likelihood given $S = t$. In particular neither g_w nor f_w is divided by the marginal probability of state w .

Augment A by $k - n$ independent Gaussian rows and call the result C , with A as its first n rows. Relative to the product measure $\mathbf{m}$ of the Gaussian law of C , $\sigma(ds)$, and counting measure on the two state sets, the real law P and the comparison measure Q on (C, S, V, W) have densities

$$\begin{aligned} \frac{dP}{d\mathbf{m}}(C, s, v, w) &= f_v(s) h_{vw}(A, As), \\ \frac{dQ}{d\mathbf{m}}(C, s, v, w) &= p_C(Cs) \bar{f}_v(C, Cs) h_{vw}(A, As). \end{aligned} \quad (97)$$

The factor $p_C(Cs)$ turns the conditional fiber average $\bar{f}_v(C, Cs)$ into the density at Cs of the projection of $f_v d\sigma$. This unnormalized equal-label measure will yield the inverse-distance kernel in Lemma 7.2. The routing weights in Q remain the actual weights $h_{vw}(A, As)$. We have $P \ll Q$: $p_C(Cs) > 0$ almost surely, and a fiber on which $\bar{f}_v(C, Cs) = 0$ has $f_v = 0$ almost everywhere on that fiber.

The comparison measure has strictly positive finite mass. Indeed, summing the routing probabilities and then the old-state likelihoods gives

$$m_Q := Q(\Omega) = \mathbb{E}_C \int p_C(Cs) d\sigma(s) = \mathbb{E}_C \int p_C(z)^2 dz \leq \mathbb{E}_C \sup_z p_C(z) < \infty.$$

The last expectation is finite by (92). Gram–Schmidt on the Gaussian rows expresses $\det(CC^\top)^{1/2}$ as a product of successive perpendicular lengths with conditional chi laws of degrees $d, d-1, \dots, d-k+1$. A chi variable of degree $\nu > 1$ has finite inverse first moment $2^{-1/2} \Gamma((\nu-1)/2) / \Gamma(\nu/2)$, and all these degrees are greater than one. Strict positivity follows from the positivity of p_C on the interior of its support.

Use the signed finite-reference entropy $D_{\text{fin}}(P||Q) = \mathbb{E}_P \log(dP/dQ)$ from Lemma 2.2. Cancellation in (97), together with the independence of C from (S, V) , yields

$$D_{\text{fin}}(P||Q) = \Phi(V) + \mathbb{E}[-\log p_C(CS)] \leq \Phi(V) + \frac{k}{2} \log(2\pi e). \quad (98)$$

Here $\log p_C(CS)$ is integrable: the logarithm of the determinant has finite log moments by the same chi factorization, and the residual factor in (92) has finite log moments by the beta law. For the inequality, let φ_k be the standard Gaussian density. Nonnegativity of $D(p_C dz || \varphi_k dz)$, integrated over C , gives $\mathbb{E}[-\log p_C(CS)] \leq \mathbb{E}[-\log \varphi_k(CS)]$. Unconditionally CS is standard Gaussian, because S has unit length and is independent of the Gaussian rows. This assertion is not a conditional Gaussian assertion given C or the state.

It is necessary to normalize Q before invoking probability data processing. Set $Q_0 = Q/m_Q$, and let π be any measurable map. Both Q and its pushforward have mass m_Q , so

$$\begin{aligned} D_{\text{fin}}(P\|Q) &= D(P\|Q_0) - \log m_Q, \\ D_{\text{fin}}(\pi_{\#}P\|\pi_{\#}Q) &= D(\pi_{\#}P\|\pi_{\#}Q_0) - \log m_Q \\ &\leq D(P\|Q_0) - \log m_Q = D_{\text{fin}}(P\|Q). \end{aligned} \quad (99)$$

No nonnegativity of $D_{\text{fin}}(P\|Q)$ is used.

Let $q_w(s)$ be the density of the (S, W) marginal of Q relative to σ and counting measure. The corresponding real density is $f_w(s)$. Equations (98) and (99) imply

$$\mathbb{E} \log \frac{f_W(S)}{q_W(S)} \leq \Phi(V) + Kd, \quad (100)$$

where this expectation, and all state expectations below, are under the real law. The densities q_w need not sum to a probability density.

We next calculate this marginal in a form suitable for geometric estimates. Let γ_b denote the law of b unrestricted standard Gaussian rows. For a unit vector u , let $\gamma_{b, \perp u}$ be the law of b independent rows with Gaussian covariance $I - uu^T$, supported on $u^\perp$.

Lemma 7.2 (Two-point Gaussian fiber measure). *Assume $1 \leq k < d - 1$. For every bounded Borel function $H(C, s, t)$ on $\mathbb{R}^{k \times d} \times S^{d-1} \times S^{d-1}$,*

$$\begin{aligned} &\int \gamma_k(dC) \int \sigma(ds) p_C(Cs) \int \sigma_{C, Cs}(dt) H(C, s, t) \\ &= (2\pi)^{-k/2} \int \sigma(ds) \int \sigma(dt) \|s - t\|^{-k} \int \gamma_{k, \perp \widehat{s-t}}(dC) H(C, s, t), \end{aligned} \quad (101)$$

where $\widehat{s-t} = (s-t)/\|s-t\|$ off the diagonal. The diagonal has product measure zero, and its assigned value is immaterial. Both sides define finite Borel measures.

Proof. For $\delta > 0$, let $\kappa_\delta(z) = (2\pi\delta^2)^{-k/2} e^{-\|z\|^2/(2\delta^2)}$. First let H be bounded and continuous, and integrate

$$H(C, s, t) \kappa_\delta(C(t-s)) \quad \text{against} \quad \gamma_k(dC) \sigma(ds) \sigma(dt).$$

Fix full-rank C and s with $\|P_C s\| < 1$. Disintegration in $z = Ct$ makes the integral in t

$$\int \kappa_\delta(z - Cs) p_C(z) \left(\int H(C, s, t) d\sigma_{C, z}(t) \right) dz.$$

The function multiplied by κ_δ is continuous at Cs . Indeed the displayed fiber parametrization is continuous at interior observations, and the test is bounded and continuous. It is bounded in absolute value by $\|H\|_\infty \sup_z p_C(z)$. The Gaussian approximate identity therefore converges to the left integrand in (101); the preceding inverse determinant estimate makes this bound integrable in C .

Alternatively fix $s \neq t$, and integrate first in C . Writing $v = t - s$, each row integral has total weight

$$(2\pi)^{-1/2} (\|v\|^2 + \delta^2)^{-1/2}.$$

After division by that weight, the row is a centered Gaussian with covariance $I - vv^T/(\|v\|^2 + \delta^2)$. As $\delta \downarrow 0$ this law converges to the Gaussian law on $v^\perp$. The k rows remain independent, so the limit is the right integrand in (101). Its absolute value is dominated by $(2\pi)^{-k/2} \|H\|_\infty \|s - t\|^{-k}$. This

is integrable under $\sigma \otimes \sigma$: dyadic annuli about s , with (90) for the small radii, give a convergent geometric series because $k < d - 1$.

The two orders compute the same limit. They prove equality on bounded continuous tests of two finite Borel measures on the indicated Polish space. Uniqueness of finite Borel measures then proves the equality for bounded Borel tests as well. This last step is what permits measurable routing functions in the application. $\square$

On the fiber $Ct = Cs$, the first n rows give $At = As$. Consequently, using (96) without normalizing its weights,

$$q_w(s) = \mathbb{E}_C \left[p_C(Cs) \int g_w(A, t) d\sigma_{C, Cs}(t) \right].$$

Apply Lemma 7.2 to $H(C, s, t) = \psi(s)g_w(A, t)$ for bounded Borel ψ . The extra $k - n$ rows do not enter g_w , so integrating them out leaves precisely the marginal law of the n actual rows. We obtain, for σ -almost every s ,

$$q_w(s) = (2\pi)^{-k/2} \int \|s - t\|^{-k} \mathbb{E}_{A \sim \gamma_{n, \perp s-t}} g_w(A, t) d\sigma(t). \quad (102)$$

This is an equality of marginal densities of the finite measure Q ; there is no additional normalization on its right-hand side.

7.3 Two estimates for a common local mass

Put $a_w = \int f_w d\sigma = \Pr\{W = w\}$, and omit states with $a_w = 0$. Define

$$L_{w,r}(s) = r^{-k} \int_{B(s,r)} f_w(t) d\sigma(t), \quad L_w(s) = \sup_{r>0} L_{w,r}(s). \quad (103)$$

For each r the integral is Borel in s . Rational radii suffice in the supremum: approximate any radius from above and use continuity from above of the finite measure $f_w\sigma$. Thus L_w is Borel. The diameter-two ball gives its lower bound, while (90) for $r \leq 1/2$ and total mass at most one for larger radii give the upper bound

$$2^{-k} a_w \leq L_w(s) \leq 2^k. \quad (104)$$

Choose the state-count-dependent cutoff

$$J = 1 + \left\lceil \frac{\log_2 N}{d-1-k} \right\rceil, \quad r_* = 2^{-J}. \quad (105)$$

Then $r_* \leq 1/2$, $Nr_*^{d-1-k} \leq 1$, and $J = O(d)$, since $d-1-k \geq d/2$ for large d and $\log N \leq d^2$.

Lemma 7.3 (Comparison with an independent projection). *Let C' have k independent standard Gaussian rows and be independent of (S, W) . Then*

$$\mathbb{E} \frac{L_W(S)}{f_W(C', C'S)} \leq e^{Kd}. \quad (106)$$

Proof. Fix a full-rank C' and a radius $r \leq 2$. Choose a finite maximal r -separated set $\{z_i\}$ on the sphere, and partition the sphere into Borel cells $E_i \subseteq B(z_i, r)$, breaking ties by their indices. The enlarged balls $B(z_i, 2r)$ overlap at most 5^d times: balls of radius $r/2$ about their separated centers have disjoint interiors, and all those meeting a given point in an enlarged ball lie inside the ambient ball of radius $5r/2$ about that point.

On E_i ,

$$L_{w,r}(s) \leq r^{-k} \int_{B(z_i, 2r)} f_w d\sigma.$$

Write R for the row space of C' , p_R for the density in (91), and σ_x for the corresponding uniform fiber law. Conditioning on $P_{C'}S = x$ gives

$$\begin{aligned} \int_{E_i} \frac{f_w(s)}{\bar{f}_w(C', C's)} d\sigma(s) &= \int_R p_R(x) \frac{\int_{E_i} f_w(s) d\sigma_x(s)}{\bar{f}_w(C', C'x)} dx \\ &\leq \Pr\{P_{C'}S \in B(P_{C'}z_i, r)\} \leq e^{Kd} r^k. \end{aligned}$$

The quotient is defined to be zero on zero-denominator fibers; on all other fibers its numerator is at most its denominator. The numerator vanishes on fibers outside the displayed projected ball. The last probability is under the unweighted prior and is bounded by (93). Multiplying the last two estimates and summing the cells, with their 5^d overlap, yields

$$\int \frac{f_w(s) L_{w,r}(s)}{\bar{f}_w(C', C's)} d\sigma(s) \leq e^{Kd} a_w. \quad (107)$$

For $r \geq r_*$, one of the grid radii $2, 1, 1/2, \dots, r_*$ gives $L_{w,r} \leq 2^k L_{w,\text{grid}}$; radii greater than two are already bounded by the radius-two expression. For $r < r_*$, (90) gives $L_{w,r} \leq r^{d-1-k} \leq r_*^{d-1-k}$. A second conditioning on the projection gives $\int f_w / \bar{f}_w(C', C' \cdot) d\sigma \leq 1$. We may therefore sum (107) over the grid and over states, obtaining

$$\sum_w \int \frac{f_w L_w}{\bar{f}_w(C', C' \cdot)} d\sigma \leq 2^k (J+2) e^{Kd} \sum_w a_w + N r_*^{d-1-k} \leq e^{Kd}.$$

The bound is uniform in full-rank C' , and averaging proves the claim. $\square$

The other comparison is with the actual finite-measure marginal:

$$\mathbb{E} \frac{q_W(S)}{L_W(S)} \leq e^{Kd}. \quad (108)$$

We prove it now. For each positive-mass state introduce the finite measure

$$d\nu_w(s) = \frac{f_w(s)}{L_w(s)} d\sigma(s).$$

Equation (104) gives $\nu_w(S^{d-1}) \leq 2^k$. More locally, for every ambient center z and $r > 0$,

$$\nu_w(B(z, r)) \leq (2r)^k. \quad (109)$$

Indeed, if $m = \int_{B(z,r)} f_w d\sigma > 0$, every s in that ball on the sphere has $L_w(s) \geq (2r)^{-k} m$; integration of f_w / L_w over the ball gives the bound. The case $m = 0$ is immediate.

Inserting (102) shows that the left side of (108) is $\sum_w \int q_w d\nu_w$. First take the portion with $\|s - t\| \leq r_*$. Uniformly in s , dyadic annuli and (90) give

$$\int_{\{\|s-t\| \leq r_*\}} \|s - t\|^{-k} d\sigma(t) \leq \frac{2^k r_*^{d-1-k}}{1 - 2^{-(d-1-k)}}. \quad (110)$$

Using $g_w \leq 1$, summing the at most N masses $\nu_w(S^{d-1}) \leq 2^k$, and retaining the prefactor in (102), this portion is at most

$$(2\pi)^{-k/2} \frac{N 2^{2k} r_*^{d-1-k}}{1 - 2^{-(d-1-k)}} \leq e^{Kd}.$$

Thus the state count in the near singular part is paid by the cutoff chosen in (105).

For the remaining distances use the annuli $r/2 < \|s - t\| \leq r$ with outer radii $r = 2, 1, \dots, 2r_*$. Fix w, t, r , and push $r^{-k}\nu_w$ restricted to that annulus to the unit direction $u = (s - t)/\|s - t\|$; call the resulting measure μ . It has total mass at most 2^k by (109). For a linear subspace E of dimension $0 \leq j \leq \ell - 1$ and $0 < \eta \leq 1$, its directions satisfy

$$\mu\{u : \text{dist}(u, E) \leq \eta\} \leq 3^j 4^k \eta^{k-j}. \quad (111)$$

In fact the corresponding s 's lie within distance ηr of the affine subspace $t + E$, with their projections there in the radius- r ball about t . That j -ball has an ηr -net of at most $(3/\eta)^j$ points by ambient packing. Balls of radius $2\eta r$ about these points cover the s 's. Applying (109) and multiplying by r^{-k} proves (111).

Lemma 7.4 (Averaging perpendicular Gaussian laws). *Let μ be a finite measure on S^{d-1} of mass at most 2^k satisfying (111) for every linear subspace of dimension $j \leq \ell - 1$. Define the finite measure on $n \times d$ matrices*

$$\lambda = \int \gamma_{n, \perp u} d\mu(u).$$

This is a Borel measure: a perpendicular matrix can be realized as $Z(I - uu^T)$ with $Z \sim \gamma_n$. Then every bounded Borel function H on this matrix space satisfies

$$\left| \int H d\lambda \right| \leq e^{Kd} \|H\|_{L^{\ell/(\ell-1)}(\gamma_n)}. \quad (112)$$

Proof. For E of dimension $j \leq \ell - 1$, layer-cake integration of (111) gives the inverse moment

$$\int \text{dist}(u, E)^{-n} d\mu(u) \leq 2^k + \frac{n 3^j 4^k}{k - j - n} \leq e^{Kd}. \quad (113)$$

Here $\text{dist}(u, E) \leq 1$; its layer-cake integral below one is $n \int_0^1 \eta^{-n-1} \mu\{\text{dist}(u, E) \leq \eta\} d\eta$. The denominator is positive because $k - j - n \geq k - \ell + 1 - n \geq d/10$.

For $u_1, \dots, u_\ell$, let $G = (\langle u_i, u_j \rangle)_{i,j \leq \ell}$. The tube estimate, sent to $\eta \downarrow 0$, shows that each proper span of at most $\ell - 1$ preceding directions has μ -mass zero. Thus G is invertible for $\mu^{\otimes \ell}$ -almost every tuple. Gram-Schmidt gives

$$\det G = \prod_{i=1}^{\ell} \text{dist}(u_i, \text{span}(u_1, \dots, u_{i-1}))^2.$$

Integrate $u_\ell, u_{\ell-1}, \dots, u_1$ in that order and apply (113) at each step. This proves

$$\int (\det G)^{-n/2} \prod_{i=1}^{\ell} d\mu(u_i) \leq e^{Kd\ell}. \quad (114)$$

It remains to pass from this determinant moment to singular row laws. For $0 < \delta \leq 1$, set

$$\alpha_\delta = \Pr\{|Z| \leq \delta\}, \quad Z \sim N(0, 1),$$

$$D_\delta(A) = \int \frac{\mathbf{1}_{\{\|Au\|_\infty \leq \delta\}}}{\alpha_\delta^n} d\mu(u).$$

There is an absolute $c > 0$ such that $\alpha_\delta \geq c\delta$. Expanding the integer ℓ -th moment under γ_n produces one tuple $u_1, \dots, u_\ell$. For an invertible Gram matrix G , the projections of each unrestricted Gaussian

row onto those directions have density at most $(2\pi)^{-\ell/2}(\det G)^{-1/2}$. The box of allowed projections has volume $(2\delta)^\ell$. Row independence and (114) therefore give

$$\mathbb{E}_{\gamma_n} D_\delta^\ell \leq \left(\frac{(2\delta)^\ell (2\pi)^{-\ell/2}}{\alpha_\delta^\ell} \right)^n e^{Kd\ell} \leq e^{Kd\ell}.$$

The null set of dependent tuples was removed above. Thus $\|D_\delta\|_{L^\ell(\gamma_n)} \leq e^{Kd}$ uniformly in δ .

The measure $D_\delta d\gamma_n$ is the mixture, with mixing measure μ , of the laws conditioned on $\|Au\|_\infty \leq \delta$. For each u , this conditioning preserves the independent perpendicular components and makes all components along u tend to zero. For every bounded continuous test H , bounded convergence in u consequently gives $\int H D_\delta d\gamma_n \rightarrow \int H d\lambda$. Hölder's inequality proves (112) for those tests. Both γ_n and λ are finite Euclidean Borel measures. Bounded continuous functions are dense in $L^{\ell/(\ell-1)}(\gamma_n + \lambda)$; approximation in this space passes the inequality to every bounded Borel H . In particular no absolute continuity of the singular mixture λ was assumed before the dual estimate. $\square$

For the far-annulus contribution to $\sum_w \int q_w d\nu_w$, apply Lemma 7.4 to the annular measure above and $H(A) = g_w(A, t)$. Since $0 \leq g_w \leq 1$,

$$\int \mathbb{E}_{\gamma_n, \perp u} g_w(A, t) d\mu(u) \leq e^{Kd} (\mathbb{E}_{\gamma_n} g_w(A, t))^{1-1/\ell} = e^{Kd} f_w(t)^{1-1/\ell}.$$

On its annulus, $\|s - t\|^{-k} \leq 2^k r^{-k}$. The definition of μ thus bounds the annular integration in s in (102) by $2^k e^{Kd} f_w(t)^{1-1/\ell}$, apart from the displayed kernel prefactor. Concavity gives, pointwise in t ,

$$\sum_w f_w(t)^{1-1/\ell} \leq N^{1/\ell} \left(\sum_w f_w(t) \right)^{1-1/\ell} \leq N^{1/\ell} \leq e^{Kd}.$$

There are $J + 1 = O(d)$ far annuli. Summing them and integrating in t , then adding the near contribution, proves (108).

7.4 Logarithmic combination

We can now finish Proposition 7.1. All logarithms in the following split are integrable under the real (S, W) law and the independent C' . Indeed

$$\mathbb{E}[-\log f_W(S)] = H(W | S) \leq \log N, \quad \mathbb{E}[-\log \tilde{f}_W(C', C'S)] = H(W | C', C'S) \leq \log N.$$

Also (104) and $\mathbb{E}[-\log a_W] = H(W) \leq \log N$ make $\log L_W(S)$ integrable. Finally $P_{S,W} \ll Q_{S,W}$. For any finite measure Q' and probability $P' \ll Q'$, the negative part of its log density ratio has bound

$$\int_{\{r < 1\}} r \log(1/r) dQ' \leq Q'(\Omega)/e, \quad r = \frac{dP'}{dQ'}.$$

Applied to the marginal comparison, this and the finite upper bound (100) make $\log(f_W(S)/q_W(S))$ integrable. Hence $\log q_W(S)$ is integrable as well, and $q_W(S) > 0$ almost surely under the real law.

Using the fresh C' in the new-state potential, we may therefore write

$$\begin{aligned}
\Phi(W) &= \mathbb{E} \log \frac{f_W(S)}{q_W(S)} + \mathbb{E} \log \frac{q_W(S)}{L_W(S)} \\
&\quad + \mathbb{E} \log \frac{L_W(S)}{\bar{f}_W(C', C'S)} \\
&\leq \Phi(V) + Kd + \log \mathbb{E} \frac{q_W(S)}{L_W(S)} \\
&\quad + \log \mathbb{E} \frac{L_W(S)}{\bar{f}_W(C', C'S)} \leq \Phi(V) + Kd.
\end{aligned}$$

The first inequality uses (100) and Jensen; the last uses (108) and Lemma 7.3. This proves the proposition. The extra $k - n$ rows allowed both local comparisons to use exponent k , whereas only the n actual rows entered the perpendicular mixture moment.

Subsection 8.6 applies the estimate to a learner and combines it with stopping and terminal accuracy to obtain a sample lower bound.

8 From row comparisons to streaming lower bounds

To apply the row comparisons to the learner in Definition 1.1, we first replace completed-measurable rules by Borel versions in the selected uniform-prior experiment. We then account for the stopping index and quantify the information that an accurate output must retain after independent auxiliary rows are revealed. These common reductions leave each comparison its own block estimate, row dimensions, and terminal residual-radius bound.

8.1 Borel rules after the shared seed is fixed

Lemma 1.2 is applied to the original jointly measurable experiment. It supplies a fixed seed for which (1) holds and the required uniform-prior success is retained. The following reduction is performed only after that choice. It does not assert preservation of behavior at every individual signal.

Lemma 8.1 (Borel versions of the finite-state rules). *Fix a seed satisfying (1), a dimension $d \geq 2$, and a deterministic finite horizon T . Retain the conditional law of U_0 at that seed. Suppose each transition and stopping probability at a positive sample index is measurable in the completion of the Borel sigma field on $\mathbb{R}^d \times \mathbb{R}$ under*

$$\lambda(dx, dy) = \gamma_1(dx) dy.$$

There are everywhere defined Borel transition and stopping kernels with the same state sets and terminal output kernels whose fixed-seed uniform-prior joint law of the signal, state path, stopping index, and output is the same as before. Their block routing probabilities are Borel functions of the entering state and the entire block data, including at states and data that have zero probability in the original run.

Proof. For each state and positive index, combine the next-state and stopping choices into one finite destination set, tagging a destination as continuing or terminal. The kernel is a finite vector of completed-measurable nonnegative functions of (x, y) whose sum is one. Every completed-measurable real function has a Borel version: approximate it by simple functions and replace their level sets by Borel sets modulo null sets. Choose such a version for every coordinate. The union of the

exceptional sets is contained in a Borel λ -null set. On the Borel set where the chosen vector is not a probability vector, replace it by a fixed point mass. That set is also null. Since the horizon and all state sets are finite, a single Borel λ -null set Z contains every sample-pair modification. A stopping choice at index zero depends only on the initial state and the fixed seed, with fresh randomness, and has no sample argument to replace.

In the fixed-seed law, (1) leaves $S \sim \sigma$ independent of the pre-generated Gaussian rows. For a nonzero row x , the image of σ under $s \mapsto \langle x, s \rangle$ has a Lebesgue density when $d \geq 2$. This follows from the one-coordinate sphere formula after rotation and multiplication by $\|x\|$. The zero row has Gaussian probability zero. Hence every marginal sample pair $(X_t, \langle X_t, S \rangle)$ is absolutely continuous with respect to λ . In particular,

$$\mathbb{P}\{(X_t, \langle X_t, S \rangle) \in Z\} = 0.$$

For any state u , even one selected using previous observations, the probability that the state before time t is u and the current pair lies in Z is bounded by this same zero probability. A finite union over the pre-generated indices therefore shows that no modified input is visited almost surely.

Couple the original and Borel finite kernels with the same fresh uniform random variable at each step. Induction makes their state paths and stopping choices identical almost surely. At identical terminal state-index pairs their output kernels agree, so the outputs may also be coupled identically. This proves the claimed joint-law equality. For any specified entering state, finite composition of the everywhere Borel kernels gives Borel block routing probabilities on every block-data vector. Such a routing function is a continuation from the specified state; it need not be a conditional law given that the state was reached. $\square$

The order of these reductions matters. Original joint measurability makes the conditional success function of the shared seed measurable, so it can be averaged in Lemma 1.2. Lemma 8.1 then chooses versions separately for the selected seed. No jointly measurable selection of replacement versions across seed values is needed. In particular the global Borel extensions used in the finite fiber comparison are fixed before its singular comparison measure is constructed; no real-law null set is transferred to a perpendicular Gaussian law.

In each fixed-seed experiment below, unused future rows are still independent of the joint vector consisting of the signal and the current state. This follows from the pre-generated product row law, (1), and the fresh transition randomness in the model. If stopping is allowed at index zero, let $\bar{U}_0$ denote the initial padded state, including that status. It is generated from U_0 and fresh time-zero randomness, so

$$\bar{U}_0 \perp\!\!\!\perp (S, (X_t)_{t \geq 1})$$

in the fixed-seed law. Whenever an auxiliary matrix is introduced, it is sampled independently of the pair consisting of S and the state in question. These joint independence statements justify the zero initial potentials and the fresh-block comparisons.

8.2 Common stopping and terminal estimates

The five comparisons use different information potentials. Their application to a learner shares two estimates. First we account for the stopping index before any analytic comparison is invoked. We then bound the information that success requires after auxiliary rows have been revealed.

Lemma 8.2 (Padding, output capacity, and unrestricted data). *Consider a fixed-seed Borel learner from Lemmas 1.2 and 8.1. Suppose its accuracy satisfies $0 < \epsilon \leq 1/10$, its persistent states have*

at most 2^M values, and its deterministic finite horizon is T . Write $L = \log(1/\epsilon)$. It admits an absorbing extension, including to the end of a longer final block, with at most

$$N = (T + 2)2^M$$

values at each layer. The terminal output is a kernel of the extended state alone. The initial extended state is independent of the joint signal and row sequence. At every deterministic block boundary, unused future rows are independent of the pair consisting of the signal and the entering extended state. The block transition is a Borel kernel of that state and the exact block data.

If its uniform-prior angular success probability is p , then

$$p \leq (T + 1)2^M \epsilon^{d-1}. \quad (115)$$

For a sequence of such learners with $p \geq 3/5$ and $M(d) = o(d^2)$, on the branch $T \leq dL$,

$$dL = O(M + 1), \quad L = o(d), \quad \log N = M \log 2 + \log(T + 2) = o(d^2). \quad (116)$$

All these estimates are uniform over the allowed accuracies. For all sufficiently large d , even a learner given all its exact observations must have $T > d/4$ to achieve success at least $3/5$.

Proof. After stopping, retain the terminal state together with its stopping index in an absorbing state. Defer the output until the end of the extension and apply its original state-index kernel. Active states and all stopped pairs number at most $(T + 2)2^M$. At index zero the stopping status is generated from the data-independent initialization and fresh randomness. The independence statements then follow from the pre-generated product row law. Finite composition of the Borel rules gives the asserted block kernel, including on inputs never visited in the original run.

There are at most $(T + 1)2^M$ terminal state-index labels. For each label, its output kernel has no further dependence on the signal. In the success integral for that label, bound the conditional probability of reaching it by one. Lemma 2.1, averaged over the output kernel, bounds the remaining integral by ϵ^{d-1} . Summing proves (115), including for randomized output. When $p \geq 3/5$, taking logarithms gives

$$(d - 1)L \leq M \log 2 + \log(T + 1) + \log(5/3).$$

For sufficiently large d , uniformly for $L \geq \log 10$, $(d - 1)L \geq 3dL/4$ and $\log(dL + 1) \leq dL/4$. On $T \leq dL$, subtraction gives $dL = O(M + 1)$. Consequently $L = o(d)$, $T = o(d^2)$, and $\log N = o(d^2)$. This is a conclusion only on the stated branch; no restriction on the accuracy sequence has been imposed.

For the unrestricted-data bound, suppose $T \leq d/4$ and grant the output all T pre-generated rows and labels, including unused pairs after a stop. Given those data, the remaining signal is uniform on a sphere of radius R in a $(d - T)$ -dimensional affine space. Indeed the row-space component is determined, while rotational invariance leaves a uniform direction in the kernel. More generally, if U is the span of $t < d$ independent Gaussian rows, isotropy and independence give

$$\mathbb{E}\|P_U S\|^2 = t/d. \quad (117)$$

Taking $t = T$ gives $\mathbb{E}(1 - R^2) = T/d$, so

$$\Pr\{R < 1/2\} \leq \frac{4T}{3d} \leq \frac{1}{3}.$$

On the complementary event, an output ball of radius ϵ intersecting that residual sphere is contained in a ball of radius 2ϵ centered at a residual-sphere point. Rescaling by $R \geq 1/2$ and applying

Lemma 2.1 bounds conditional success by $(4\epsilon)^{d-T-1}$. The learner's remaining randomness has no further signal information after all data are given, so the same estimate holds after averaging its output. Thus success is at most

$$\frac{1}{3} + (2/5)^{3d/4-1} < \frac{3}{5}$$

for large d , a contradiction. This argument includes $T = 0$, when the residual sphere is the original unit sphere. $\square$

Lemma 8.3 (Residual information in a successful state). *Let $S \sim \sigma$, let V be a finite state jointly distributed with S , and let G have $0 \leq k < d - 1$ independent standard Gaussian rows, independently of (S, V) . Suppose a unit-vector output is drawn from a kernel of V alone and has angular success probability at least p at accuracy $\epsilon > 0$. Let R be the radius of the conditional sphere of S given (G, GS) . If*

$$0 < \rho \leq 1, \quad 2\epsilon < \rho, \quad \Pr\{R < \rho\} \leq \eta < p,$$

then

$$I(S; V \mid G, GS) \geq (p - \eta)(d - k - 1) \log \frac{\rho}{2\epsilon} - \log 2. \quad (118)$$

For $\rho < 1$, one available bad-radius bound is

$$\Pr\{R < \rho\} \leq \frac{k}{d(1 - \rho^2)}. \quad (119)$$

Proof. Let Q preserve the side-data marginal (G, GS) and the conditional marginals of S, V given those data, while making S and V conditionally independent. Apply the same state-output kernel under both laws. Their relative entropy, including the output, is $I(S; V \mid G, GS)$.

The event of angular success and $R \geq \rho$ has actual probability at least $p - \eta$. Under Q , condition also on the output. The residual signal remains uniform on its sphere in the kernel of G , independently of the output. Angular success implies Euclidean error at most ϵ . If the corresponding output ball intersects the residual sphere, centering it at one point of the intersection enlarges its radius to at most 2ϵ . Rescaling the residual sphere to unit radius and applying Lemma 2.1 bounds the conditional success probability by $(2\epsilon/\rho)^{d-k-1}$.

For any event with actual probability a and reference probability at most b , where $0 < b \leq 1$, binary relative entropy and data processing give

$$D(P\|Q) \geq a \log(1/b) - \log 2. \quad (120)$$

Indeed the binary entropy of a is at most $\log 2$. Applying this inequality to the preceding event proves (118). Finally (117) gives $\mathbb{E}(1 - R^2) = k/d$; Markov's inequality proves (119). $\square$

We now apply the comparisons to these extended states. The critical-radius, density-level, and hybrid comparisons use an entropy bound for the outgoing state; the fiber comparison uses its number of values. The same width bound supplies both hypotheses. Each route retains its own auxiliary dimension and terminal residual-sphere estimate.

8.3 The critical-radius comparison

Theorem 8.4 (Streaming consequence of the critical radius). *Let $M(d)$ be a nonnegative integer sequence with $M(d) = o(d^2)$, and let $0 < \epsilon(d) \leq 1/10$. For each d , consider a learner in Definition 1.1*

with deterministic finite horizon $T(d)$. If its uniform-sphere angular success probability, including all learner randomness, is at least $3/5$, then

$$T(d) \geq c d \log \frac{1}{\epsilon(d)}$$

for an absolute $c > 0$ and all sufficiently large d . The eventual dimension threshold may depend on the memory sequence.

Proof. Use Lemma 1.2 with $p_0 = 3/5$, and then Lemma 8.1. The resulting fixed-seed uniform-prior experiment retains success at least $3/5$, the conditional initial-state law, the same state bound, and the same horizon. Transition and terminal output randomness remain fresh.

Put $L = \log(1/\epsilon) \geq \log 10$. The branch $T \geq dL$ already has the desired order. On $T < dL$, Lemma 8.2 gives padded states of entropy at most $H_* = M \log 2 + \log(T + 2) \leq d^2$ eventually. In particular its output-capacity estimate specializes to

$$\frac{3}{5} \leq (T + 1) 2^M \epsilon^{d-1}. \quad (121)$$

Use $B = \lceil T/m \rceil$ full padded blocks, where $m = \lfloor d/10 \rfloor$ as in (18).

For a boundary state U , define

$$J(U) = I(S; U \mid G, GS)$$

with an independent ℓ -row matrix G , $\ell = \lfloor d/2 \rfloor$, sampled independently of the pair (S, U) . In a block, write U, W for the entering and exiting states and A for its m rows. The matrix A is independent of (S, U) . Append an independent $r = \ell - m$ row matrix C and set $G = (A; C)$. In the aligned experiment (G, GS) reveals the entire exact block data (A, AS) . Conditional on these data and U , the block transition has no further dependence on S . Conditional data processing therefore gives

$$I_1(S; W \mid G, GS) \leq I_1(S; U \mid G, GS) = J(U).$$

The equality uses independence of G from the pair (S, U) . Theorem 4.1, applied to the outgoing state with $H(W) \leq H_* \leq d^2$, yields

$$J(W) \leq J(U) + Kd.$$

The initial padded state $\bar{U}_0$ is independent of S , and its auxiliary matrix is independent of $(S, \bar{U}_0)$, so $J(\bar{U}_0) = 0$. Iteration gives for the final padded state W_f

$$J(W_f) \leq Kd \left\lceil \frac{T}{m} \right\rceil. \quad (122)$$

We next lower-bound that final information. Conditional on (G, GS) , the row-space projection of S is determined and the residual direction is uniform in $\ker G$. If $a = d - \ell$, its radius R has the marginal representation

$$R^2 \stackrel{\text{law}}{=} \frac{X}{X + Y}, \quad X \sim \chi_a^2, \quad Y \sim \chi_\ell^2$$

with independent X, Y . This follows by normalizing a standard Gaussian vector. Since $a \geq d/2$, the events $X \geq 0.75a$ and $X + Y \leq 1.4d$ imply $R \geq 1/2$. Chebyshev's inequality, using variances $2a$ and $2d$, gives

$$\mathbb{P}\{R < 1/2\} \leq \frac{32}{a} + \frac{25}{2d} = O(1/d). \quad (123)$$

For sufficiently large d , the radius tail in (123) leaves success probability at least $1/2$ on $R \geq 1/2$. Apply Lemma 8.3 with $\rho = 1/2$ and that tail bound. It gives

$$J(W_f) \geq \frac{a-1}{2}(L - \log 4) - \log 2 \geq c_0 dL \quad (124)$$

for an absolute $c_0 > 0$ and all sufficiently large d . This is uniform over $L \geq \log 10$, because

$$L - \log 4 \geq \left(1 - \frac{\log 4}{\log 10}\right) L > 0.$$

The common stopping reduction gives $T > d/4 \geq m$, so $\lceil T/m \rceil \leq 2T/m$. Equation (122) is therefore an $O(T)$ upper bound. Comparing it with (124) proves $T \geq cdL$ uniformly over the stated accuracies. $\square$

8.4 The density-level comparisons

We apply both comparisons to Definition 1.1. Assume that the original uniform-prior experiment has angular success probability at least $2/3$, that $0 < \epsilon \leq 1/10$, and that $M = M(d) = o(d^2)$; T is its deterministic finite horizon. Put $L = \log(1/\epsilon) \geq \log 10$. A worst-signal success premise of $2/3$ first implies this uniform-prior premise by integration.

Apply Lemma 1.2 to the original jointly measurable experiment with $p_0 = 2/3$. Fix a seed ξ for which (1) holds and conditional success is at least $2/3$, retain $\mathcal{L}(U_0 \mid \Xi = \xi)$, and then apply Lemma 8.1. These reductions preserve the fixed-seed uniform-prior joint law, the state bound, and the horizon. All learner statements in this subsection concern that experiment. We prove $T \geq cdL$ for an absolute $c > 0$ in all sufficiently large dimensions; the eventual threshold may depend on the memory sequence.

For completeness, (120) applied to success under the product signal/output law and Lemma 2.1 give the unconditional bound

$$I(S; \hat{S}) \geq \frac{2}{3}(d-1)L - \log 2. \quad (125)$$

Restrict to $T < dL$. Lemma 8.2 provides absorbing padded states, a state-only terminal output, data-independent initial state $\bar{U}_0$, and fresh future blocks. Its clock bound implies, in particular,

$$L \leq C \frac{M+1+\log d}{d}, \quad T = O(M+d), \quad H_* := M \log 2 + \log(T+2) = o(d^2). \quad (126)$$

The lemma also gives

$$T > d/4. \quad (127)$$

Both comparisons below use this padded learner. Their auxiliary matrices and block lengths remain distinct.

For the one-level comparison, return to (32) and use $B_{\text{one}} = \lceil T/k \rceil$ padded blocks. In a block let Z and U be the entering and exiting states. The next matrix X is independent of the pair (S, Z) . Draw B independently of (S, Z, X, U) , and put $G = (X; B)$ as in Proposition 5.3. Conditional on (G, GS) , the entire block data (X, XS) are known. The transition kernel from Z to U then has no additional dependence on S . Conditional data processing gives

$$\tilde{\mathcal{D}}(U) \leq I_{\text{act}}(S; Z \mid G, GS) = \mathcal{D}(Z),$$

where the equality uses independence of G from (S, Z) . By (126), $H(U) \leq H_* \leq d^2$ eventually, so Proposition 5.3 costs at most Cd per block. The initial boundary $\bar{U}_0$ is independent of S , and its auxiliary G is independent of the pair $(S, \bar{U}_0)$; hence $\mathcal{D}(\bar{U}_0) = 0$, and

$$\mathcal{D}(U_f) \leq Cd B_{\text{one}} \quad (128)$$

for the final padded state.

Take the independent m -row matrix in the definition of $\mathcal{D}$. Its residual radius R satisfies

$$\Pr\{R < 1/2\} \leq \frac{m/d}{3/4} \leq \frac{1}{3}.$$

Lemma 8.3, with $p = 2/3$, $\rho = 1/2$, and $\eta = 1/3$, now gives

$$\mathcal{D}(U_f) \geq \frac{d-m-1}{3}(L - \log 4) - \log 2 \geq c_0 dL \quad (129)$$

for an absolute $c_0 > 0$ in sufficiently large dimension. The final inequality is uniform for $L \geq \log 10 > \log 4$. Since $T > d/4 \geq k$, the common raw-data bound makes $\lceil T/k \rceil \leq 2T/k$. Hence (128) is $O(T)$, and comparison with (129) proves $T \geq cdL$.

For the two-label comparison, use (45) and $B_{\text{two}} = \lceil T/n \rceil$ padded batches. Let V_i be the state after i batches. At a batch the incoming A is independent of (S, V_i) , and B is independent of the whole actual experiment. Apply Proposition 5.4 to the outgoing state V_{i+1} ; the required entropy bound follows from (126) on the branch $T < dL$. Given (D, DS) , the exact batch (A, AS) is known, so the update is conditionally a channel from V_i without further access to S . Therefore

$$\mathcal{J}(V_{i+1}) \leq I_{\text{act}}(S; V_{i+1} \mid D, DS) + Cd \leq I_{\text{act}}(S; V_i \mid D, DS) + Cd = \mathcal{J}(V_i) + Cd.$$

The last equality uses independence of D from (S, V_i) . Set $V_0 = \bar{U}_0$. It is independent of S , and its auxiliary G is independent of the pair (S, V_0) ; hence $\mathcal{J}(V_0) = 0$. Iteration gives

$$\mathcal{J}(V_{B_{\text{two}}}) \leq Cd B_{\text{two}} \leq C'(T + d). \quad (130)$$

For the independent k -row matrix defining $\mathcal{J}$, (119) at $\rho = 1/4$ gives

$$\Pr\{R < 1/4\} \leq \frac{16k}{15d} \leq \frac{8}{15}.$$

Apply Lemma 8.3 with $p = 2/3$ and $\eta = 8/15$. It gives

$$\mathcal{J}(V_{B_{\text{two}}}) \geq \frac{2}{15}(d-k-1)(L - \log 8) - \log 2. \quad (131)$$

This is at least cdL uniformly for $L \geq \log 10$, since $L - \log 8 \geq (1 - \log 8 / \log 10)L > 0$ and $d-k-1 \geq d/3$. The upper bound (130) is $O(T)$ by $T > d/4$. This proves $T \geq cdL$; the discarded branch $T \geq dL$ already has the same order.

In both arguments the auxiliary rows are analysis variables. The learner receives only its prescribed fresh Gaussian stream with exact labels, and the comparisons permit arbitrary computation within a transition subject to the persistent finite-state restriction.

8.5 The hybrid comparison: stopping and terminal information

The averaged block estimate combines with the common stopping reduction. We need a terminal information bound uniform over the auxiliary row counts in that average.

Lemma 8.5 (Information in a successful terminal state). *Let V be a padded final state whose unit-vector output has uniform-prior angular success at least $2/3$ at $0 < \epsilon \leq 1/10$. For every $k_0 \leq k \leq k_1$,*

$$J_k(V) \geq \frac{d-k-1}{14} \log \frac{1}{5\epsilon} - \log 2. \quad (132)$$

Proof. For every $k_0 \leq k \leq k_1$, the auxiliary matrix has $k \leq d/2$ rows. At $\rho = 2/5$, (119) therefore gives

$$\Pr\{R < 2/5\} \leq \frac{k/d}{1 - 4/25} \leq \frac{25}{42}.$$

The success probability is at least $2/3$, and $2/3 - 25/42 = 1/14$. Lemma 8.3 now gives (132), because $\rho/(2\epsilon) = 1/(5\epsilon)$. Its radius condition holds for $\epsilon \leq 1/10$. $\square$

Theorem 8.6 (Precision lower bound from hybrid projections). *Let $M(d)$ be a nonnegative integer sequence with $M(d) = o(d^2)$, and let $0 < \epsilon(d) \leq 1/10$. For each d , consider a learner in Definition 1.1 with memory bound $M(d)$, accuracy $\epsilon(d)$, deterministic horizon $T(d) \geq 0$, and uniform prior on S^{d-1} . If*

$$\mathbb{P}\{\angle(\hat{S}, S) \leq \epsilon(d)\} \geq \frac{2}{3},$$

then

$$T(d) \geq cd \log \frac{1}{\epsilon(d)}$$

for all sufficiently large d . The constant $c > 0$ is absolute; the dimension threshold may depend on the memory sequence.

Proof. Apply Lemma 1.2 with $p_0 = 2/3$ to the original jointly measurable experiment. Fix a seed satisfying (1) with conditional uniform-prior success at least $2/3$, retain its conditional initial-state law, and then apply Lemma 8.1. The fixed-seed joint experiment, state bound, and horizon are preserved.

Put $b = \log(1/\epsilon)$. In dimensions with $T > db$, the conclusion already holds. In the remaining dimensions, Lemma 8.2 permits padding with $W = (T + 2)2^M$ states and $B = \log W = o(d^2)$, so eventually $B \leq d^2$. Break the padded computation into blocks of h_0 observations, with a shorter final block if necessary. The initial padded state $\bar{U}_0$ is independent of S , and the side matrix is independent of the pair $(S, \bar{U}_0)$. Thus $J_k(\bar{U}_0) = 0$ for every k , and its averaged potential is zero. Proposition 6.1 gives

$$\bar{J}(V_{\text{final}}) \leq Cd \left\lceil \frac{T}{h_0} \right\rceil \leq C'(T + d) \leq C''T.$$

The last inequality uses the independent bound $T > d/4$ from Lemma 8.2; thus it also covers a shorter last block.

Lemma 8.5 gives the opposite bound for every k in the average. Since $k \leq d/2$ and $b \geq \log 10$,

$$\log \frac{1}{5\epsilon} = b - \log 5 \geq \frac{\log 2}{\log 10} b.$$

The right side of (132) is consequently at least $c'db$ for all sufficiently large d , uniformly over the allowed accuracies. Comparing the two bounds proves $T \geq cdb$. The dimensions with $T > db$ were already covered. $\square$

If a learner succeeds for every unit signal with probability at least $2/3$, it also satisfies the uniform-prior hypothesis, so the theorem applies to that worst-signal guarantee. The shared-seed case is covered by the joint-independence reduction at the start of the proof, which retains fresh randomized transition and output kernels. The proof uses exact labels and imposes no linearity or within-transition precision restriction.

8.6 The finite Gaussian-fiber comparison

We apply Proposition 7.1 to Definition 1.1, including its common notation $\Xi, U_0, (X_t)_{t \geq 1}$. This route also fixes the remaining randomness after choosing the seed, so its terminal count uses a deterministic decoder.

Corollary 8.7 (Uniform-sphere sample lower bound). *Let $M(d) \geq 0$ be integer-valued with $M(d) = o(d^2)$, and let $0 < \epsilon(d) \leq 1/10$. Suppose learners in Definition 1.1, with $S \sim \sigma$, satisfy*

$$\Pr\{\arccos\langle \hat{S}, S \rangle \leq \epsilon(d)\} \geq 2/3,$$

where the probability includes the signal, rows, and learner randomness. Then

$$T(d) \geq c d \log(1/\epsilon(d))$$

for an absolute $c > 0$ and all sufficiently large d . The eventual dimension threshold may depend on the memory sequence.

Proof. Apply Lemma 1.2 with $p_0 = 2/3$ to the original jointly measurable uniform-prior experiment. Fix a seed ξ satisfying (1) whose conditional success is at least $2/3$, and retain the conditional initial-state law. Then apply Lemma 8.1 in that fixed-seed experiment. In particular all global Borel routing extensions are fixed before (96) and the finite measure Q are formed. The two-point identity applies to those Borel functions without transferring a null set from the real observation law to a perpendicular row law.

On a product extension of this fixed-seed experiment, choose an array $\mathcal{R}$ of independent uniform random variables consisting of R_t , $0 \leq t \leq T$, and $R_{t,j}^{\text{out}}$, $0 \leq t \leq T$, $1 \leq j \leq 2^M$. Choose the whole array independently of $(U_0, S, (X_t)_{t \geq 1})$. The factorization (1) then implies that the complete conditional choice vector $(U_0, \mathcal{R})$ is independent of $(S, (X_t)_{t \geq 1})$.

Enumerate each finite state alphabet by slots $j \leq 2^M$. Use R_0 for a stopping choice before the first sample and, at time $t \geq 1$, use R_t to realize the joint next-state and stopping outcome by the inverse distribution function of its finite probability row. Use $R_{t,j}^{\text{out}}$ to realize the unit-vector output kernel at terminal pair (t, j) . Such a realization can be chosen Borel for a probability kernel on the standard Borel sphere, by mapping the sphere into a Borel subset of $[0, 1]$ and using an inverse distribution function. The resulting fixed-seed experiment is jointly measurable and has the same conditional law as the Borel learner.

Its success probability averaged over $(U_0, \mathcal{R})$ is at least $2/3$. Fubini and averaging give a complete realization with uniform-prior success at least $3/5$. Fix it. The initial state, all transition and stopping rules, and all terminal outputs are now deterministic. Independence of the complete choice vector leaves the joint law of the signal and all rows unchanged. Thus the complete vector $(\xi, U_0, \mathcal{R})$, including the seed and initialization, is fixed before any state comparison. Its unused entries add no persistent learner state.

Put $X_* = d \log(1/\epsilon)$, and restrict to $T < X_*$. Apply Lemma 8.2 to the fixed decoder with success at least $3/5$. Its output count gives

$$(d-1) \log(1/\epsilon) \leq M \log 2 + \log(T+1) + \log(5/3). \quad (133)$$

The common short-run estimate yields $X_* = O(M + 1)$, and the absorbing padded state has width at most

$$\begin{aligned} N &= (T + 2)2^M, \\ \log N &\leq M \log 2 + \log(X_* + 2) = O(M + 1) \leq d^2 \end{aligned} \tag{134}$$

for all sufficiently large d . The padded decoder is a function of this state, and rows generated after stopping remain fresh.

Pad also to $\lceil T/n \rceil$ full n -row blocks. Each block's row matrix is independent of the pair consisting of the signal and old state, while its labels remain exact. After the complete fixation, the initial padded state is deterministic, so its potential is zero. Iterating Proposition 7.1 for the padded final state F gives

$$\Phi(F) \leq Kd\lceil T/n \rceil. \tag{135}$$

For a lower bound use the independent k -row matrix in $\Phi(F)$. Equation (119) gives

$$\Pr\{R < 1/2\} \leq \frac{4k}{3d} \leq \frac{4}{9}.$$

The fixed decoder has success at least $3/5$. Apply Lemma 8.3 with $p = 3/5$, $\rho = 1/2$, and $\eta = 4/9$, and write $p_0 = 3/5 - 4/9 > 0$. It gives

$$\begin{aligned} \Phi(F) &\geq p_0(d - k - 1) \log(1/(4\epsilon)) - \log 2 \\ &\geq cd \log(1/\epsilon) \end{aligned} \tag{136}$$

for sufficiently large d . The constant is absolute because $d - k - 1 \geq d/2$ and $\log(1/(4\epsilon)) \geq (1 - \log 4 / \log 10) \log(1/\epsilon)$ on the stated accuracy range.

Finally, Lemma 8.2 gives $T > d/4 \geq n$, including the exclusion of $T = 0$. Hence $\lceil T/n \rceil \leq 2T/n$, and d/n is absolutely bounded for large d . The upper bound (135) is therefore $O(T)$. Combining it with (136) proves the corollary. $\square$

If a learner has angular success at least $2/3$ for every fixed signal, averaging that guarantee over the uniform sphere supplies the hypothesis of Corollary 8.7. It therefore gives the same lower bound with the stated worst-signal premise.

References

- [1] Tim Austin. Multi-variate correlation and mixtures of product measures. *Kybernetika*, 56(3):459–499, 2020. [doi:10.14736/kyb-2020-3-0459](https://doi.org/10.14736/kyb-2020-3-0459).
- [2] Yuval Dagan, Gil Kur, and Ohad Shamir. Space lower bounds for linear prediction in the streaming model. In *Proceedings of the Thirty-Second Conference on Learning Theory*, Proceedings of Machine Learning Research 99, pages 929–954, 2019. <https://proceedings.mlr.press/v99/dagan19b.html>. Theorem 3 and Theorem 10 locators refer to this published PMLR version.
- [3] Paul Dupuis and Yixiang Mao. Formulation and properties of a divergence used to compare probability measures without absolute continuity. *ESAIM: Control, Optimisation and Calculus of Variations*, 28, Article 10, 2022. [doi:10.1051/cocv/2022002](https://doi.org/10.1051/cocv/2022002).
- [4] Pertti Mattila. Hausdorff dimension, orthogonal projections and intersections with planes. *Annales Academiae Scientiarum Fennicae, Series A I, Mathematica*, 1(2):227–244, 1975. [doi:10.5186/aasfm.1975.0110](https://doi.org/10.5186/aasfm.1975.0110).

- [5] OpenAI. Localization costs and information growth for exact Gaussian observations. OpenAI Math Release preprint [OAI:Localization-costs-and-information-growth-for-exact-Gaussian-observations-September-27-2026](#), 2026.
- [6] Ran Raz. Fast learning requires good memory: A time-space lower bound for parity learning. arXiv:1602.05161v1, 16 February 2016. <https://arxiv.org/abs/1602.05161v1>.
- [7] Ran Raz. A time-space lower bound for a large class of learning problems. In *Proceedings of the 58th Annual IEEE Symposium on Foundations of Computer Science*, pages 732–742, 2017. doi:10.1109/FOCS.2017.73. Theorem and section locators refer to ECCC Report TR17-020, 12 February 2017: <https://eccc.weizmann.ac.il/report/2017/020/>.
- [8] Daniel Russo and James Zou. Controlling bias in adaptive data analysis using information theory. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research 51, pages 1232–1240, 2016. <https://proceedings.mlr.press/v51/russo16.html>.
- [9] Vatsal Sharan, Aaron Sidford, and Gregory Valiant. Memory-sample tradeoffs for linear regression with small error. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pages 890–901, 2019. doi:10.1145/3313276.3316403. Theorem and section locators refer to the full version arXiv:1904.08544v2, 10 October 2020: <https://arxiv.org/abs/1904.08544v2>.
- [10] Jacob Steinhardt and John Duchi. Minimax rates for memory-bounded sparse linear regression. In *Proceedings of the 28th Conference on Learning Theory*, Proceedings of Machine Learning Research 40, pages 1564–1587, 2015. <https://proceedings.mlr.press/v40/Steinhardt15.html>.
- [11] Tim van Erven and Peter Harremoës. Rényi divergence and Kullback–Leibler divergence. arXiv:1206.2459v2, 24 April 2014. <https://arxiv.org/abs/1206.2459v2>. Theorem and equation locators refer to this version.
- [12] Aolin Xu and Maxim Raginsky. Information-theoretic analysis of generalization capability of learning algorithms. In *Advances in Neural Information Processing Systems* 30, pages 2524–2533, 2017. https://papers.nips.cc/paper_files/paper/2017/hash/ad71c82b22f4f65b9398f76d8be4c615-Abstract.html.